

27. Realtime Clock (RTC)

27.1 Overview

The realtime clock (RTC) has two counting modes, calendar count mode and binary count mode, that are used by switching register settings. For calendar count mode, the RTC has a 100-year calendar from 2000 to 2099 and automatically adjusts dates for leap years. For binary count mode, the RTC counts seconds and retains the information as a serial value. Binary count mode can be used for calendars other than the Gregorian (Western) calendar.

The sub-clock oscillator or LOCO can be selected as the count source of the time counters. The RTC uses a 128-Hz clock acquired by dividing the count source by a prescaler. Year, month, date, day-of-week, a.m. /p.m. (in 12-hour mode), hour, minute, second, or 32-bit binary is counted by 1/128 second.

Table 27.1 lists the RTC specifications, Figure 27.1 shows a block diagram, and Table 27.2 lists the I/O pins.

Table 27.1 RTC specifications

Parameter	Specifications
Count mode	Calendar count mode/binary count mode
Count source ^{*1}	Sub-clock (XCIN), External clock input (EXCIN) or LOCO
Clock and calendar functions	- Calendar count mode - Year, month, date, day of week, hour, minute, second are counted, BCD display 12 hours/24 hours mode switching function 30 seconds adjustment function (a number less than 30 is rounded down to 00 seconds, and 30 seconds or more are rounded up to 1 minute) - Automatic adjustment function for leap years - Binary count mode - Count seconds in 32 bits, binary display - Shared by both modes - Start/stop function - The sub-second digit is displayed in binary units (1 Hz, 2 Hz, 4 Hz, 8 Hz, 16 Hz, 32 Hz, or 64 Hz) - Clock error correction function - Clock (1-Hz/64-Hz) output
Interrupts	- Alarm interrupt (RTC_ALM) - As an alarm interrupt condition, selectable for comparison with the following: - Calendar count mode: Year, month, date, day-of-week, hour, minute, or second can be selected - Binary count mode: Each bit of the 32-bit binary counter - Periodic interrupt (RTC_PRD) 2 seconds, 1 second, 1/2 second, 1/4 second, 1/8 second, 1/16 second, 1/32 second, 1/64 second, 1/128 second, or 1/256 second can be selected as an interrupt period. - Carry interrupt (RTC_CUP) - An interrupt is generated at either of the following conditions: - When a carry from the 64-Hz counter to the second counter is generated. - When the 64-Hz counter is changed and the R64CNT register is read at the same time. (32-KHz count mode is only for 64-Hz counter reading) - Return from Software Standby or Deep Software Standby mode can be performed by the alarm interrupt or periodic interrupt
Time capture function	- Times can be captured when the edge of the time capture event input pin is detected. For every event input, month, date, hour, minute, and second are captured or the 32-bit binary counter value is captured. - Interrupt can be generated when the edge of the time capture event input is detected. The time capture event input pin and IRQ are shared.
Event link function	Periodic event output (RTC_PRD)
TrustZone filter	Security and Privilege attribution can be set

Note 1. The frequency of the peripheral module clock (PCLKB) $\geq$ the frequency of the count source should be satisfied.

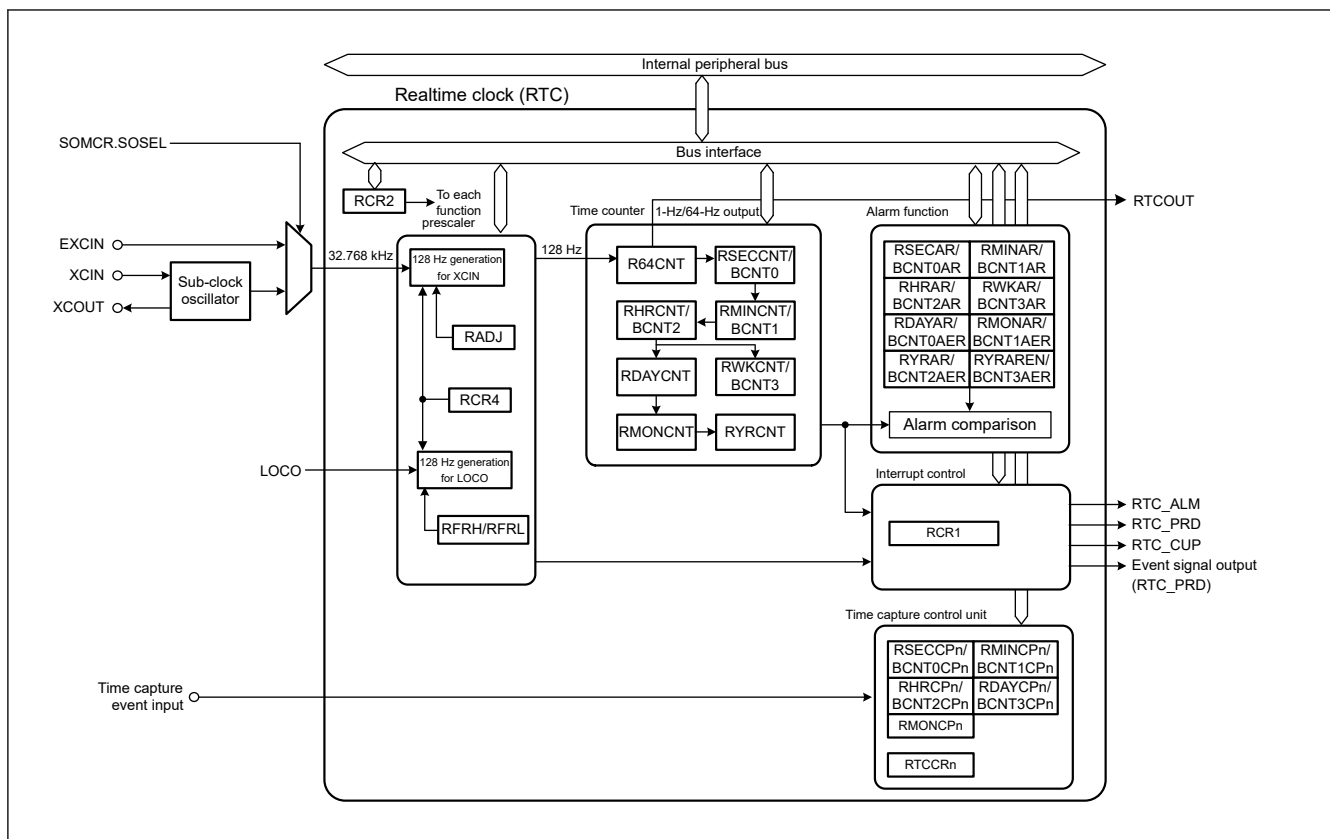


Figure 27.1 RTC block diagram

Table 27.2 RTC I/O pins

Pin name	I/O	Description
XCIN	Input	Connect a 32.768-kHz crystal to these pins
XCO	Output	
EXCIN	Input	External sub-clock input
RTCOUT	Output	This pin is used to output a 1-Hz/64-Hz waveform, but not in Deep Software Standby mode
RTCI _{Cn} (n = 0 to 2)	Input	Time capture event input pins RTCI _{Cn} can be controlled by the VBTICTLR register. For more information, see section 12, Battery Backup Function and section 21, I/O Ports .

27.2 Register Descriptions

Write or read from the RTC registers as described in [section 27.6.5. Notes on Writing to and Reading from Registers](#).

If the value in an RTC register after a reset is given as x (undefined bits) in the list, it is not initialized by a reset. When RTC enters the reset state or a low power state during counting operations, for example, while the RCR2.START bit is 1, the year, month, day of the week, date, hours, minutes, seconds, and 64-Hz counters continue to operate.

Note: A reset generated while writing to a register might destroy the register value. In addition, do not allow the MCU to enter Software Standby mode, Deep Software Standby mode, or battery backup state immediately after setting any of these registers. For details, see [section 27.6.4. Transitions to Low Power Modes After Setting Registers](#).

27.2.1 R64CNT : 64-Hz Counter

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x00

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	F1HZ	F2HZ	F4HZ	F8HZ	F16HZ	F32HZ	F64HZ

Value after reset: 0 x x x x x x x

Bit	Symbol	Function	R/W
0	F64HZ	64-Hz Flag This bit indicates the 64-Hz state of the sub-second digit.	R
1	F32HZ	32-Hz Flag This bit indicates the 32-Hz state of the sub-second digit.	R
2	F16HZ	16-Hz Flag This bit indicates the 16-Hz state of the sub-second digit.	R
3	F8HZ	8-Hz Flag This bit indicates the 8-Hz state of the sub-second digit.	R
4	F4HZ	4-Hz Flag This bit indicates the 4-Hz state of the sub-second digit.	R
5	F2HZ	2-Hz Flag This bit indicates the 2-Hz state of the sub-second digit.	R
6	F1HZ	1-Hz Flag This bit indicates the 1-Hz state of the sub-second digit.	R
7	—	This bit is read as 0.	R

Note: S-TYPE3, P-TYPE3

The R64CNT counter is used in both calendar count mode and binary count mode. The 64-Hz counter (R64CNT) generates the period for a second by counting up periods of the 128-Hz clock. The state in the sub-second range can be confirmed by reading this counter.

This counter is set to 0x00 by an RTC software reset or an execution of a 30-second adjustment. To read this counter, follow the procedure in [section 27.3.5. Reading 64-Hz Counter and Time](#).

27.2.2 RSECCNT : Second Counter (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x02

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	SEC10[2:0]			SEC1[3:0]			

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	SEC1[3:0]	1-Second Count Counts from 0 to 9 every second. When a carry is generated, 1 is added to the tens place.	R/W
6:4	SEC10[2:0]	10-Second Count Counts from 0 to 5 for 60-second counting.	R/W
7	—	The read value is undefined. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

The RSECCNT counter sets and counts the BCD-coded second value. It counts the carries generated once per second in the 64-Hz counter.

The setting range is decimal 00 to 59. The RTC does not operate normally if any other value is set. Before writing to this register, you must stop the count operation using the START bit in RCR2.

To read this counter, follow the procedure in [section 27.3.5. Reading 64-Hz Counter and Time](#).

27.2.3 RMINCNT : Minute Counter (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x04

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	MIN10[2:0]			MIN1[3:0]			
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
3:0	MIN1[3:0]	1-Minute Count Counts from 0 to 9 every minute. When a carry is generated, 1 is added to the tens place.	R/W
6:4	MIN10[2:0]	10-Minute Count Counts from 0 to 5 for 60-minute counting.	R/W
7	—	The read value is undefined. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

The RMINCNT counter sets and counts the BCD-coded minute value. It counts the carries generated once every minute in the second counter.

A value from 00 through 59 (in BCD) can be specified. If a value outside of this range is specified, the RTC does not operate correctly. Before writing to this register, you must stop the count operation using the START bit in RCR2. To read this counter, follow the procedure in [section 27.3.5. Reading 64-Hz Counter and Time](#).

27.2.4 RHRCNT : Hour Counter (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x06

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	PM	HR10[1:0]		HR1[3:0]			
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
3:0	HR1[3:0]	1-Hour Count Counts from 0 to 9 once per hour. When a carry is generated, 1 is added to the tens place.	R/W
5:4	HR10[1:0]	10-Hour Count Counts from 0 to 2 once per carry from the ones place.	R/W
6	PM	AM/PM select for time counter setting. 0: AM 1: PM	R/W
7	—	The read value is undefined. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

The RHRCNT counter sets and counts the BCD-coded hour value. It counts the carries generated once per hour in the minute counter. The specifiable time differs based on the setting in the hours mode bit (RCR2.HR24):

- When the RCR2.HR24 bit is 0 – from 00 to 11 (in BCD).
- When the RCR2.HR24 bit is 1 – from 00 to 23 (in BCD).

If a value outside of this range is specified, the RTC does not operate correctly. Before writing to this register, you must stop the count operation using the START bit in RCR2. The PM bit is only enabled when the RCR2.HR24 bit is 0.

Otherwise, the setting in the PM bit has no effect. To read this counter, follow the procedure in [section 27.3.5. Reading 64-Hz Counter and Time](#).

27.2.5 RWKCNT : Day-of-Week Counter (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x08

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	DAYW[2:0]		
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
2:0	DAYW[2:0]	Day-of-Week Counting 0 0 0: Sunday 0 0 1: Monday 0 1 0: Tuesday 0 1 1: Wednesday 1 0 0: Thursday 1 0 1: Friday 1 1 0: Saturday 1 1 1: Setting prohibited	R/W
7:3	—	The read values are undefined. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

The RWKCNT counter sets and counts in the coded day-of-week value. It counts the carries generated once per day in the hour counter. A value from 0 through 6 can be specified. If a value outside of this range is specified, the RTC does not operate correctly. Before writing to this register, you must stop the count operation using the START bit in RCR2. To read this counter, follow the procedure in [section 27.3.5. Reading 64-Hz Counter and Time](#).

27.2.6 BCNTn : Binary Counter n (n = 0 to 3) (in Binary Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x02 + 0x02 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	BCNT[7:0]							
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
7:0	BCNT[7:0]	Binary Counter	R/W

Note: S-TYPE3, P-TYPE3

BCNTn is a read/write 8-bit register to access BCNT[31:0] that is a 32-bit binary counter. BCNT3 is assigned to the BCNT[31:24] bits, BCNT2 is assigned to the BCNT[23:16] bits, BCNT1 is assigned to the BCNT[15:8] bits, and BCNT0 is assigned to the BCNT[7:0] bits. BCNTn performs count operation by a carry generated for each second of the 64-Hz counter. Before writing to this register, you must stop the count operation using the START bit in RCR2. To read this counter, follow the procedure in [section 27.3.5. Reading 64-Hz Counter and Time](#).

27.2.7 RDAYCNT : Day Counter

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x0A

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	DATE10[1:0]	DATE1[3:0]				

Value after reset: 0 0 x x x x x x

Bit	Symbol	Function	R/W
3:0	DATE1[3:0]	1-Day Count Counts from 0 to 9 once per day. When a carry is generated, 1 is added to the tens place.	R/W
5:4	DATE10[1:0]	10-Day Count Counts from 0 to 3 once per carry from the ones place.	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

The RDAYCNT counter is used in calendar count mode to set and count the BCD-coded date value. It counts the carries generated once per day in the hour counter. The count operation depends on the month and whether the year is a leap year. Leap years are determined according to whether the year counter (RYRCNT) value is divisible by 400, 100, and 4.

A value from 01 through 31 (in BCD) can be specified. If a value outside of this range is specified, the RTC does not operate correctly. When specifying a value, the range of specifiable days depends on the month and whether the year is a leap year. Before writing to this register, you must stop the count operation using the START bit in RCR2. To read this counter, follow the procedure in [section 27.3.5. Reading 64-Hz Counter and Time](#).

27.2.8 RMONCNT : Month Counter

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x0C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	MON10	MON1[3:0]			

Value after reset: 0 0 0 x x x x x

Bit	Symbol	Function	R/W
3:0	MON1[3:0]	1-Month Count Counts from 0 to 9 once per month. When a carry is generated, 1 is added to the tens place.	R/W
4	MON10	10-Month Count Counts from 0 to 1 once per carry from the ones place.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

The RMONCNT counter is used in calendar count mode to set and count the BCD-coded month value. It counts the carries generated once per month in the date counter.

A value from 01 through 12 (in BCD) can be specified. If a value outside of this range is specified, the RTC does not operate correctly. Before writing to this register, you must stop the count operation using the START bit in RCR2. To read this counter, follow the procedure in [section 27.3.5. Reading 64-Hz Counter and Time](#).

27.2.9 RYRCNT : Year Counter

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x0E

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	YR10[3:0]				YR1[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
3:0	YR1[3:0]	1-Year Count Counts from 0 to 9 once per year. When a carry is generated, 1 is added to the tens place.	R/W
7:4	YR10[3:0]	10-Year Count Counts from 0 to 9 once per carry from ones place. When a carry is generated in the tens place, 1 is added to the hundreds place.	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

The RYRCNT counter is used in calendar count mode to set and count the BCD-coded year value. It counts the carries generated once per year in the month counter.

A value from 00 through 99 (in BCD) can be specified. If a value outside of this range is specified, the RTC does not operate correctly. Before writing to this register, you must stop the count operation using the START bit in RCR2. To read this counter, follow the procedure in [section 27.3.5. Reading 64-Hz Counter and Time](#).

27.2.10 RSECAR : Second Alarm Register (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x10

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ENB	SEC10[2:0]			SEC1[3:0]			
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
3:0	SEC1[3:0]	1 Second Value for the ones place of seconds.	R/W
6:4	SEC10[2:0]	10 Seconds Value for the tens place of seconds.	R/W
7	ENB	ENB 0: Do not compare register value with RSECCNT counter value 1: Compare register value with RSECCNT counter value	R/W

Note: S-TYPE3, P-TYPE3

RSECAR is an alarm register associated with the BCD-coded second counter RSECCNT. When the ENB bit is set to 1, the RSECAR value is compared with the RSECCNT value. From the following alarm registers, only those selected with the ENB bits set to 1 are compared with the associated counters:

- RSECAR
- RMINAR
- RHRAR
- RWKAR
- RDAYAR

- RMONAR
- RYRAREN

When all the respective values match, the IR flag associated with the RTC_ALM interrupt is set to 1. RSECAR values from 00 through 59 (in BCD) can be specified. If a value outside of this range is specified, the RTC does not operate correctly. This register is set to 0x00 by an RTC software reset.

27.2.11 RMINAR : Minute Alarm Register (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x12

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ENB	MIN10[2:0]			MIN1[3:0]			

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	MIN1[3:0]	1 Minute Value for the ones place of minutes.	R/W
6:4	MIN10[2:0]	10 Minutes Value for the tens place of minutes.	R/W
7	ENB	ENB 0: Do not compare register value with RMINCNT counter value 1: Compare register value with RMINCNT counter value	R/W

Note: S-TYPE3, P-TYPE3

RMINAR is an alarm register associated with the BCD-coded minute counter RMINCNT. When the ENB bit is set to 1, the RMINAR value is compared with the RMINCNT value. From the following alarm registers, only those selected with the ENB bits set to 1 are compared with the associated counters:

- RSECAR
- RMINAR
- RHRAR
- RWKAR
- RDAYAR
- RMONAR
- RYRAREN

When all the respective values match, the IR flag associated with the RTC_ALM interrupt is set to 1. RMINAR values from 00 through 59 (in BCD) can be specified. If a value outside of this range is specified, the RTC does not operate correctly. This register is set to 0x00 by an RTC software reset.

27.2.12 RHRAR : Hour Alarm Register (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x14

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ENB	PM	HR10[1:0]		HR1[3:0]			

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	HR1[3:0]	1 Hour Value for the ones place of hours.	R/W
5:4	HR10[1:0]	10 Hours Value for the tens place of hours.	R/W
6	PM	AM/PM select for alarm setting. 0: AM 1: PM	R/W
7	ENB	ENB 0: Do not compare register value with RHRCNT counter value 1: Compare register value with RHRCNT counter value	R/W

Note: S-TYPE3, P-TYPE3

RHRAR is an alarm register associated with the BCD-coded hour counter RHRCNT. When the ENB bit is set to 1, the RHRAR value is compared with the RHRCNT value. From the following alarm registers, only those selected with the ENB bits set to 1 are compared with the associated counters:

- RSECAR
- RMINAR
- RHRAR
- RWKAR
- RDAYAR
- RMONAR
- RYRAREN

When all the respective values match, the IR flag associated with the RTC_ALM interrupt is set to 1. The specifiable time differs according to the setting in the hours mode bit (RCR2.HR24):

- When the RCR2.HR24 bit is 0 – From 00 to 11 (in BCD).
- When the RCR2.HR24 bit is 1 – From 00 to 23 (in BCD).

If a value outside of this range is specified, the RTC does not operate correctly. When the RCR2.HR24 bit is 0, you must set the PM bit. When the RCR2.HR24 bit is 1, the setting in the PM bit has no effect. This register is set to 0x00 by an RTC software reset.

27.2.13 RWKAR : Day-of-Week Alarm Register (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x16

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ENB	—	—	—	—	—	DAYW[2:0]	
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
2:0	DAYW[2:0]	Day-of-Week Setting 0 0 0: Sunday 0 0 1: Monday 0 1 0: Tuesday 0 1 1: Wednesday 1 0 0: Thursday 1 0 1: Friday 1 1 0: Saturday 1 1 1: Setting prohibited	R/W
6:3	—	The read values are undefined. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
7	ENB	ENB 0: Do not compare register value with RWKCNT counter value 1: Compare register value with RWKCNT counter value	R/W

Note: S-TYPE3, P-TYPE3

RWKAR is an alarm register associated with the coded day-of-week counter RWKCNT. When the ENB bit is set to 1, the RWKAR value is compared with the RWKCNT value. From the following alarm registers, only those selected with the ENB bits set to 1 are compared with the associated counters:

- RSECAR
- RMINAR
- RHRAR
- RWKAR
- RDAYAR
- RMONAR
- RYRAREN

When all the respective values match, the IR flag associated with the RTC_ALM interrupt is set to 1. RWKAR values from 0 through 6 (in BCD) can be specified. If a value outside of this range is specified, the RTC does not operate correctly. This register is set to 0x00 by an RTC software reset.

27.2.14 BCNTnAR : Binary Counter n Alarm Register (n = 0 to 3) (in Binary Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x10 + 0x02 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	BCNTAR							

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
7:0	BCNTAR	Alarm register associated with the 32-bit binary counter	R/W

Note: S-TYPE3, P-TYPE3

BCNTnAR is a read/write alarm register associated with the 32-bit binary counter. BCNT3AR is assigned to the BCNTAR[31:24] bits, BCNT2AR is assigned to the BCNTAR[23:16] bits, BCNT1AR is assigned to the BCNTAR[15:8] bits, and BCNT0AR is assigned to the BCNTAR[7:0]. This register is set to 0x00 by an RTC software reset.

27.2.15 RDAYAR : Date Alarm Register (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x18

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ENB	—	DATE10[1:0]	DATE1[3:0]				

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	DATE1[3:0]	1 Day Value for the ones place of days.	R/W

Bit	Symbol	Function	R/W
5:4	DATE10[1:0]	10 Days Value for the tens place of days.	R/W
6	—	The read value is undefined. The write value should be 0.	R/W
7	ENB	ENB 0: Do not compare register value with RDAYCNT counter value 1: Compare register value with RDAYCNT counter value	R/W

Note: S-TYPE3, P-TYPE3

RDAYAR is an alarm register associated with the BCD-coded date counter RDAYCNT. When the ENB bit is set to 1, the RDAYAR value is compared with the RDAYCNT value. From the following alarm registers, only those selected with the ENB bits set to 1 are compared with the associated counters:

- RSECAR
- RMINAR
- RHRAR
- RWKAR
- RDAYAR
- RMONAR
- RYRAREN

When all the respective values match, the IR flag associated with the RTC_ALM interrupt is set to 1. The RDAYAR values from 01 through 31 (in BCD) can be specified. If a value outside of this range is specified, the RTC does not operate correctly. This register is set to 0x00 by an RTC software reset.

27.2.16 RMONAR : Month Alarm Register (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x1A

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ENB	—	—	MON1 0	MON1[3:0]			

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
3:0	MON1[3:0]	1 Month Value for the ones place of months.	R/W
4	MON10	10 Months Value for the tens place of months.	R/W
6:5	—	The read values are undefined. The write value should be 0.	R/W
7	ENB	ENB 0: Do not compare register value with RMONCNT counter value 1: Compare register value with RMONCNT counter value	R/W

Note: S-TYPE3, P-TYPE3

RMONAR is an alarm register associated with the BCD-coded month counter RMONCNT. When the ENB bit is set to 1, the RMONAR value is compared with the RMONCNT value. From the following alarm registers, only those selected with the ENB bits set to 1 are compared with the associated counters:

- RSECAR
- RMINAR
- RHRAR
- RWKAR

- RDAYAR
- RMONAR
- RYRAREN

When all the respective values match, the IR flag associated with the RTC_ALM interrupt is set to 1. The RMONAR values from 01 through 12 (in BCD) can be specified. If a value outside of this range is specified, the RTC does not operate correctly. This register is set to 0x00 by an RTC software reset.

27.2.17 RYRAR : Year Alarm Register (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x1C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	YR10[3:0]				YR1[3:0]			
Value after reset:	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
3:0	YR1[3:0]	1 Year Value for the ones place of years.	R/W
7:4	YR10[3:0]	10 Years Value for the tens place of years.	R/W
15:8	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

RYRAR is an alarm register associated with the BCD-coded year counter RYRCNT. The RYRAR values from 00 through 99 (in BCD) can be specified. If a value outside of this range is specified, the RTC does not operate correctly. This register is set to 0x0000 by an RTC software reset.

27.2.18 RYRAREN : Year Alarm Enable Register (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x1E

Bit position:	7	6	5	4	3	2	1	0
Bit field:	ENB	—	—	—	—	—	—	—
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
6:0	—	The read values are undefined. The write value should be 0.	R/W
7	ENB	ENB 0: Do not compare register value with the RYRCNT counter value 1: Compare register value with the RYRCNT counter value	R/W

Note: S-TYPE3, P-TYPE3

When the ENB bit in the RYRAREN register is set to 1, the RYRAR value is compared with the RYRCNT value. From the following alarm registers, only those selected with the ENB bits set to 1 are compared with the associated counters:

- RSECAR
- RMINAR
- RHRAR
- RWKAR

- RDAYAR
- RMONAR
- RYRAREN

When all the respective values match, the IR flag associated with the RTC_ALM interrupt is set to 1. This register is set to 0x00 by an RTC software reset.

27.2.19 BCNTnAER : Binary Counter n Alarm Enable Register (n = 0 to 3) (in Binary Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x18 + 0x02 × n

Bit position: 7 6 5 4 3 2 1 0

Bit field: ENB[7:0]

Value after reset: x x x x x x x x

Bit	Symbol	Function	R/W
7:0	ENB[7:0]	Setting the alarm enable associated with the 32-bit binary counter	R/W

Note: S-TYPE3, P-TYPE3

BCNTnAER is a read/write register for setting the alarm enable (BCNTAER) associated with the 32-bit binary counter. BCNT3AER is assigned to the BCNTAER.ENB[31:24] bits, BCNT2AER register is assigned to the BCNTAER.ENB[23:16] bits, BCNT1AER is assigned to the BCNTAER.ENB[15:8] bits, and BCNT0AER is assigned to the BCNTAER.ENB[7:0] bits. The binary counter (BCNT[31:0]) associated with the BCNTAER.ENB[31:0] bits that are set to 1 is compared with the binary alarm register (BCNTAR) and, when all match, the IR flag associated with the RTC_ALM interrupt is set to 1. This register is set to 0x00 by an RTC software reset.

27.2.20 RCR1 : RTC Control Register 1

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x22

Bit position: 7 6 5 4 3 2 1 0

Bit field: PES[3:0]

RTCO
S

PIE

CIE

AIE

Value after reset: x x x x 0 x 0 x

Bit	Symbol	Function	R/W
0	AIE	Alarm Interrupt Enable 0: Disable alarm interrupt requests 1: Enable alarm interrupt requests	R/W
1	CIE	Carry Interrupt Enable 0: Disable carry interrupt requests 1: Enable carry interrupt requests	R/W
2	PIE	Periodic Interrupt Enable 0: Disable periodic interrupt requests 1: Enable periodic interrupt requests	R/W
3	RTCOS	RTCOUT Output Select 0: Outputs 1 Hz on RTCOUT 1: Outputs 64 Hz RTCOUT	R/W

Bit	Symbol	Function	R/W
7:4	PES[3:0]	Periodic Interrupt Select 0x6: Generate periodic interrupt every 1/256 second*1 0x7: Generate periodic interrupt every 1/128 second 0x8: Generate periodic interrupt every 1/64 second 0x9: Generate periodic interrupt every 1/32 second 0xA: Generate periodic interrupt every 1/16 second 0xB: Generate periodic interrupt every 1/8 second 0xC: Generate periodic interrupt every 1/4 second 0xD: Generate periodic interrupt every 1/2 second 0xE: Generate periodic interrupt every 1 second 0xF: Generate periodic interrupt every 2 seconds Others: Do not generate periodic interrupts	R/W

Note: S-TYPE3, P-TYPE3

Note 1. When LOCO is selected (RCR4.RCKSEL = 1) while PES[3:0] = 0x6, a periodic interrupt is generated every 1/128 second.

The RCR1 register is used in both calendar count mode and binary count mode. Bits AIE, PIE, and PES[3:0] are updated synchronously with the count source. When the RCR1 register is modified, check that all the bits are updated before proceeding.

AIE bit (Alarm Interrupt Enable)

The AIE bit enables or disables alarm interrupt requests.

If the times indicated in the counters and alarm settings match in Deep Software Standby mode, the MCU returns from Deep Software Standby mode regardless of the AIE bit value.

CIE bit (Carry Interrupt Enable)

The CIE bit enables or disables interrupt requests when a carry to the RSECCNT/BCNT0 register occurs, or when a carry to the 64-Hz counter (R64CNT) occurs while reading the 64-Hz counter.

PIE bit (Periodic Interrupt Enable)

The PIE bit enables or disabled a periodic interrupt.

If the periods indicated in the counters and PES[3:0] settings match in Deep Software Standby mode, the MCU returns from Deep Software Standby mode regardless of the PIE bit value.

RTCOS bit (RTCOUT Output Select)

The RTCOS bit selects the RTCOUT output period. The RTCOS bit must be rewritten while the count operation is stopped (RCR2.START = 0) and the RTCOUT output is disabled (RCR2.RTCOE = 0). When RTCOUT is output to an external pin, the RCR2.RTCOE bit must be enabled.

PES[3:0] bits (Periodic Interrupt Select)

The PES[3:0] bits specify the period for the periodic interrupt. A periodic interrupt is generated with the period specified in these bits.

27.2.21 RCR2 : RTC Control Register 2 (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x24

Bit position:	7	6	5	4	3	2	1	0
Bit field:	CNTM D	HR24	AADJ P	AADJ E	RTCO E	ADJ30	RESE T	START

Value after reset: x x x x 0 0 0 x

Bit	Symbol	Function	R/W
0	START	Start 0: Stop prescaler and time counter 1: Operate prescaler and time counter normally	R/W

Bit	Symbol	Function	R/W
1	RESET	RTC Software Reset 0: In writing: Invalid (writing 0 has no effect). In reading: Normal time operation in progress, or an RTC software reset has completed. 1: In writing: Initialize the prescaler and target registers for RTC software reset*1. In reading: RTC software reset in progress.	R/W
2	ADJ30	30-Second Adjustment 0: In writing: Invalid (writing 0 has no effect). In reading: Normal time operation in progress, or 30-second adjustment has completed. 1: In writing: Execute 30-second adjustment. In reading: 30-second adjustment in progress.	R/W
3	RTCOE	RTCOUT Output Enable 0: Disable RTCOUT output 1: Enable RTCOUT output	R/W
4	AADJE	Automatic Adjustment Enable*2*3 0: Disable automatic adjustment 1: Enable automatic adjustment	R/W
5	AADJP	Automatic Adjustment Period Select*2*3 0: The RADJ.ADJ[5:0] setting from the count value of the prescaler every minute. 1: The RADJ.ADJ[5:0] setting value is adjusted from the count value of the prescaler every 10 seconds.	R/W
6	HR24	Hours Mode*3 0: Operate RTC in 12-hour mode 1: Operate RTC in 24-hour mode	R/W
7	CNTMD	Count Mode Select*4 0: Calendar count mode 1: Binary count mode	R/W

Note: S-TYPE3, P-TYPE3

Note 1. R64CNT, RSECAR, RMINAR, RHRAR, RWKAR, RDAYAR, RMONAR, RYRAR, RYRAREN, RADJ, RTCCRN, RSECCPN, RMINCPN, RHRCPN, RDAYCPN, RMONCPN, RCR2.ADJ30, RCR2.AADJE, RCR2.AADJP.

Note 2. When LOCO is selected, the setting of this bit is disabled.

Note 3. When rewriting this bit, confirm that the value has been rewritten before performing the following processing. See [section 27.6.5. Notes on Writing to and Reading from Registers](#) for notes on register writing/reading.

Note 4. When rewriting this bit, confirm that the value has been rewritten before performing the following processing.

The RCR2 register is related to hours mode, automatic adjustment function, enabling RTCOUT output, 30-second adjustment, RTC software reset, and controlling count operation.

START bit (Start)

The START bit stops or restarts the prescaler or time counter operation. This bit is updated in synchronization with the next cycle of the count source. When the START bit is modified, check that the bit is updated before proceeding.

RESET bit (RTC Software Reset)

The RESET bit initializes the prescaler and registers to be reset by RTC software. When 1 is written to this bit, initialization starts in synchronization with the count source. When the initialization is complete, the RESET bit is automatically set to 0. Check that this bit is 0 before proceeding.

ADJ30 bit (30-Second Adjustment)

The ADJ30 bit is for 30-second adjustment.

When 1 is written to the ADJ30 bit, the RSECCNT value of less than 30 seconds is rounded down to 00 second and the value of 30 seconds or more is rounded up to 1 minute.

The 30-second adjustment is performed in synchronization with the count source. When 1 is written to this bit, the ADJ30 bit is automatically set to 0 after the 30-second adjustment completes. If 1 is written to the ADJ30 bit, check that the bit is 0 before proceeding. When the 30-second adjustment is performed, the prescaler and R64CNT are also reset. The ADJ30 bit is set to 0 by an RTC software reset.

RTCOE bit (RTCOUT Output Enable)

The RTCOE bit enables output of a 1-Hz/64-Hz clock signal from the RTCOUT pin.

Use the START bit to stop counting before changing the value of the RTCOE bit. Do not stop counting (write 0 to the START bit) and change the value of the RTCOE bit at the same time.

When RTCOUT is to be output from an external pin, enable the RTCOE bit and set up the port control for the pin.

AADJE bit (Automatic Adjustment Enable)

The AADJE bit controls (enables or disables) automatic adjustment.

Set the plus-minus bits (RADJ.PMADJ[1:0]) to 00b (adjustment is not performed) before changing the value of the AADJE bit.

The AADJE bit is set to 0 by an RTC software reset.

AADJP bit (Automatic Adjustment Period Select)

The AADJP bit selects the automatic-adjustment period.

Set the plus-minus bits (RADJ.PMADJ[1:0]) to 00b (adjustment is not performed) before changing the value of the AADJP bit.

The AADJP bit is set to 0 by an RTC software reset.

HR24 bit (Hours Mode)

The HR24 bit specifies whether the RTC operates in 12- or 24-hour mode.

Use the START bit to stop counting before changing the value of the HR24 bit. Do not stop counting (write 0 to the START bit) and change the value of the HR24 bit at the same time.

CNTMD bit (Count Mode Select)

The CNTMD bit specifies whether the RTC count mode operates in calendar count mode or in binary count mode.

When setting the count mode, execute an RTC software reset and start again from the initial settings. This bit is updated in synchronization with the count source. However, the count mode switches only after the RTC software reset. (Bit switches before RTC reset, mode switches after RTC reset.)

For details on initial settings, see [section 27.3.1. Outline of Initial Settings of Registers After Power On](#).

27.2.22 RCR2 : RTC Control Register 2 (in Binary Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x24

Bit position:	7	6	5	4	3	2	1	0
Bit field:	CNTM D	—	AADJ P	AADJ E	RTCO E	—	RESE T	START
Value after reset:	x	x	x	x	0	0	0	x

Bit	Symbol	Function	R/W
0	START	Start 0: Stop the 32-bit binary counter, 64-Hz counter, and prescaler 1: Operate the 32-bit binary counter, 64-Hz counter, and prescaler normally	R/W
1	RESET	RTC Software Reset 0: In writing: Invalid (writing 0 has no effect). In reading: Normal time operation in progress, or an RTC software reset has completed. 1: In writing: Initialize the prescaler and target registers for RTC software reset* ¹ . In reading: RTC software reset in progress.	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
3	RTCOE	RTCOUT Output Enable 0: Disable RTCOUT output 1: Enable RTCOUT output	R/W
4	AADJE	Automatic Adjustment Enable ^{*2*3} 0: Disable automatic adjustment 1: Enable automatic adjustment	R/W
5	AADJP	Automatic Adjustment Period Select ^{*2*3} 0: Add or subtract RADJ.ADJ [5:0] bits from prescaler count value every 32 seconds 1: Add or subtract RADJ.ADJ [5:0] bits from prescaler count value every 8 seconds.	R/W
6	—	The read value is undefined. The write value should be 0.	R/W
7	CNTMD	Count Mode Select ^{*4} 0: Calendar count mode 1: Binary count mode	R/W

Note: S-TYPE3, P-TYPE3

Note 1. R64CNT, BCNTnAR, BCNTnAER, RADJ, RTCCRN, BCNTnCpM, RCR2.ADJ30, RCR2.AADJE, RCR2.AADJP.

Note 2. When LOCO is selected, the setting of this bit is disabled.

Note 3. When rewriting this bit, confirm that the value has been rewritten before performing the following processing. See [section 27.6.5. Notes on Writing to and Reading from Registers](#) for notes on register writing/reading.

Note 4. When rewriting this bit, confirm that the value has been rewritten before performing the following processing.

RCR2 in the binary count mode is a register related to the automatic correction function, RTCOUT output enable, RTC software reset, and count mode control.

START bit (Start)

The START bit stops or restarts the prescaler or counter (clock) operation. This bit is updated in synchronization with the count source. When the START bit is modified, check that the bit is updated before proceeding.

RESET bit (RTC Software Reset)

The RESET bit initializes the prescaler and registers to be reset by RTC software. When 1 is written to this bit, initialization starts in synchronization with the count source. When the initialization is complete, the RESET bit is automatically set to 0. When 1 is written to the RESET bit, check that the bit is 0 before proceeding.

RTCOE bit (RTCOUT Output Enable)

The RTCOE bit enables output of a 1-Hz/64-Hz clock signal from the RTCOUT pin.

Use the START bit to stop counting before changing the value of the RTCOE bit. Do not stop counting (write 0 to the START bit) and change the value of the RTCOE bit at the same time. When an RTCOUT signal is to be output from an external pin, enable the port control in addition to setting this bit.

AADJE bit (Automatic Adjustment Enable)

The AADJE bit controls (enables or disables) automatic adjustment.

Set the plus-minus bits (RADJ.PMADJ[1:0]) to 00b (adjustment is not performed) before changing the value of the AADJE bit. The AADJE bit is set to 0 by an RTC software reset.

AADJP bit (Automatic Adjustment Period Select)

The AADJP bit selects the automatic-adjustment period.

Correction period can be selected from 32 second units or 8 second units in binary count mode.

Set the plus-minus bits (RADJ.PMADJ[1:0]) to 00b (adjustment is not performed) before changing the value of the AADJP bit. The AADJP bit is set to 0 by an RTC software reset.

CNTMD bit (Count Mode Select)

The CNTMD bit specifies whether the RTC count mode operates in calendar count mode or in binary count mode.

When setting the count mode, execute an RTC software reset and start again from the initial settings. This bit is updated in synchronization with the count source. However, the count mode switches only after the RTC software reset. (Bit switches before RTC reset, mode switches after RTC reset.)

For details on initial settings, see [section 27.3.1. Outline of Initial Settings of Registers After Power On](#).

27.2.23 RCR4 : RTC Control Register 4

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x28

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	RCKSEL
Value after reset:	0	0	0	0	0	0	0	x

Bit	Symbol	Function	R/W
0	RCKSEL	Count Source Select 0: Sub-clock oscillator is selected 1: LOCO is selected	R/W
7:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

The RCR4 register is used in both calendar count mode and binary count mode.

RCKSEL bit (Count Source Select)

The RCKSEL bit selects the count source from the sub-clock oscillator and LOCO.

The RCKSEL bit is only used in normal operation mode. When the RCKSEL bit is set to 0, the time is counted with the sub-clock oscillator. When the bit is set to 1, the time is counted with LOCO.

For details on count source setting, see [section 27.3.1. Outline of Initial Settings of Registers After Power On](#) and [section 27.3.2. Clock and Count Mode Setting Procedure](#). The count source must be selected only once before specifying the initial settings of the RTC registers at power on.

27.2.24 RFRL : Frequency Register L

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x2C

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RFC[15:0]															
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
15:0	RFC[15:0]	Frequency Comparison Value Write 0x00FF to this register when using the LOCO.	R/W

Note: S-TYPE3, P-TYPE3

RFRL is a register for controlling the prescaler when LOCO is selected.

The RTC time counter operates on a 128-Hz clock signal as the base clock. Therefore, when LOCO is selected, LOCO is divided by the prescaler to generate a 128-Hz clock signal. Set the frequency comparison value in the RFC[15:0] bits to generate a 128-Hz clock from the LOCO frequency. Before writing to RFC[15:0] after a cold start, write 0x0000 to the RFRH register.

A value from 0x0007 through 0x01FF can be specified as the frequency comparison value. If a value outside of this range is specified, the RTC does not operate correctly. Before writing to this register, be sure to stop the count operation through the setting of the START bit in RCR2. The operating frequency of the peripheral module clock and the LOCO should be such that the peripheral module clock is $\geq$ LOCO.

Calculation method of frequency comparison value:

$RFC[15:0] = (\text{LOCO clock frequency}) / 128 - 1$

When the LOCO frequency is 32.768 kHz, the RFRL register should be set to 0x00FF.

27.2.25 RFRH : Frequency Register H

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x2A

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RFC16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	x

Bit	Symbol	Function	R/W
0	RFC16	Write 0 before writing to the RFRL register after a cold start.	R/W
15:1	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE3, P-TYPE3

Before writing to RFRHL.RFC[15:0] after a cold start, write 0x0000 to the RFRH register.

27.2.26 RADJ : Time Error Adjustment Register

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x2E

Bit position:	7	6	5	4	3	2	1	0
Bit field:	PMADJ[1:0]		ADJ[5:0]					
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
5:0	ADJ[5:0]	Adjustment Value These bits specify the adjustment value from the prescaler.	R/W
7:6	PMADJ[1:0]	Plus-Minus 0 0: Do not perform adjustment. 0 1: Adjustment is performed by the addition to the prescaler 1 0: Adjustment is performed by the subtraction from the prescaler 1 1: Setting prohibited.	R/W

Note: S-TYPE3, P-TYPE3

The RADJ register is used in both calendar count mode and binary count mode. Adjustment is performed by the addition to or subtraction from the prescaler or 64-Hz counter. If the Automatic Adjustment Enable (RCR2.AADJE) bit is 0, adjustment is performed when writing to the RADJ. If the RCR2.AADJE bit is 1, adjustment is performed in the interval specified in the Automatic Adjustment Period Select (RCR2.AADJP) bit.

The current adjustment by software (disabling automatic adjustment) may be invalid if the following adjustment value is specified within 320 cycles of the count source after the register setting. To perform adjustment consecutively, wait for 320 cycles or more of the count source after the register setting, then specify the next adjustment value.

RADJ is updated in synchronization with the count source. When RADJ is modified, check that all the bits are updated before continuing with more processing. This register is set to 0x00 by an RTC software reset. The setting of this register is enabled only when the sub-clock oscillator is selected. When LOCO is selected, adjustment is not performed.

ADJ[5:0] bits (Adjustment Value)

The ADJ[5:0] bits specify the adjustment value (number of sub-clock cycles) from the prescaler.

PMADJ[1:0] bits (Plus-Minus)

The PMADJ[1:0] bits select whether the clock is set ahead or back depending on the error-adjustment value set in the ADJ[5:0] bits.

27.2.27 RTCCRN : Time Capture Control Register n (n = 0 to 2)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x40 + 0x02 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	TCEN	—	TCNF[1:0]	—	TCST	TCCT[1:0]		
Value after reset:	x	0	x	x	0	x	x	x

Bit	Symbol	Function	R/W
1:0	TCCT[1:0]	Time Capture Control 0 0: Do not detect events 0 1: Detect rising edge 1 0: Detect falling edge 1 1: Detect both edges	R/W
2	TCST	Time Capture Status 0: No event detected 1: Event detected ^{*1}	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
5:4	TCNF[1:0]	Time Capture Noise Filter Control 0 0: Turn noise filter off 0 1: Setting prohibited 1 0: Turn noise filter on (count source) 1 1: Turn noise filter on (count source by divided by 32)	R/W
6	—	This bit is read as 0. The write value should be 0.	R/W
7	TCEN	Time Capture Event Input Pin Enable 0: Disable the RTCICn pin as the time capture event input pin 1: Enable the RTCICn pin as the time capture event input pin	R/W

Note: S-TYPE3, P-TYPE3

Note 1. Indicates that an event is detected. Writing 1 to this bit has no effect. Writing 0 sets this bit to 0.

The RTCCRN register is used both in calendar count mode and in binary count mode. RTCCR0, RTCCR1, and RTCCR2 control the RTCIC0, RTCIC1, and RTCIC2 pins, respectively.

RTCCRN is updated in synchronization with the count source. When RTCCRN is modified, check that all the bits except the TCST bit are updated before continuing with additional processing. This register is cleared to 0x00 by an RTC software reset. When RTCICm is used as the time capture pin, VBTICTLR.VCHnIEN (n = 0 to 2) must be set to 1.

TCCT[1:0] bits (Time Capture Control)

The TCCT[1:0] bits control the edge detection of the time capture event input pins, RTCIC0, RTCIC1, and RTCIC2. The detection edge is selectable. The TCCT[1:0] bits must be set while the VBTICTLR.VCHnIEN bit is 1.

TCST bit (Time Capture Status)

The TCST bit indicates that an event on the time capture event input pins, RTCIC0, RTCIC1, and RTCIC2, was detected. When the TCST bit is 0, no event is detected. When the TCST bit is 1, this bit indicates that an event was detected on the associated pin and the capture register is valid. When multiple events are detected, the capture time for the first event is retained.

The event is detected only during count operation (RCR2.START bit = 1). Before reading the capture register, make sure that this bit is set to 1.

Set the TCST bit while the TCCT[1:0] bits are 00b (no event is detected). The TCST bit is set to 0 in synchronization with the count source. When the TCST bit is set to 0, check that the bit is updated before continuing with additional processing.

TCNF[1:0] bits (Time Capture Noise Filter Control)

The TCNF[1:0] bits control the noise filter of the time capture event input pins (RTCIC0, RTCIC1, and RTCIC2).

When the noise filter is on, the count source divided by 1 or divided by 32 is selectable. In this case, when the input level on the time capture event input pin matches three consecutive times at the set sampling period, the input level is determined.

Set the TCNF[1:0] bits while the TCCT[1:0] bits are 00b (no event is detected). When the noise filter is used, set the TCNF[1:0] bits, wait for 3 cycles of the specified sampling period, then set the TCCT[1:0] bits. Set the TCNF[1:0] bits when the VBTICTLR.VCHnIEN bit is 1.

TCEN bit (Time Capture Event Input Pin Enable)

The TCEN bit enables or disables the time capture event input pins RTCIC0, RTCIC1, and RTCIC2. When the functions of the time capture event input pins are multiplexed, set VBTICTLR first. If the TCEN bit is set to 0, also set the TCCT[1:0] bits to 00b.

27.2.28 RSECCPn : Second Capture Register n (n = 0 to 2) (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x52 + 0x10 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	SEC10[2:0]			SEC1[3:0]			
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
3:0	SEC1[3:0]	1-Second Capture Capture value for the ones place of seconds.	R
6:4	SEC10[2:0]	10-Second Capture Capture value for the tens place of seconds.	R
7	—	The read value is undefined.	R

Note: S-TYPE3, P-TYPE3

RSECCPn is a read-only register that captures the RSECCNT value when a time capture event is detected.

The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the RSECCP0, RSECCP1, and RSECCP2 registers, respectively. This register is cleared to 0x00 by an RTC software reset. Before reading from this register, the time capture event detection should be stopped using the RTCCRn.TCCT[1:0] bits.

27.2.29 RMINCPn : Minute Capture Register n (n = 0 to 2) (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x54 + 0x10 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	MIN10[2:0]			MIN1[3:0]			
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
3:0	MIN1[3:0]	1-Minute Capture Capture value for the ones place of minutes.	R
6:4	MIN10[2:0]	10-Minute Capture Capture value for the tens place of minutes.	R
7	—	The read value is undefined.	R

Note: S-TYPE3, P-TYPE3

RMINCP_n is a read-only register that captures the RMINCNT value when a time capture event is detected.

The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the RMINCP0, RMINCP1, and RMINCP2 registers, respectively.

This register is cleared to 0x00 by an RTC software reset. Before reading from this register, the time capture event detection should be stopped using the RTCCR_n.TCCT[1:0] bits.

27.2.30 RHRCP_n : Hour Capture Register n (n = 0 to 2) (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x56 + 0x10 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	PM	HR10[1:0]	HR1[3:0]				
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
3:0	HR1[3:0]	1-Hour Capture Capture value for the ones place of hours	R
5:4	HR10[1:0]	10-Hour Capture Capture value for the tens place of hours	R
6	PM	PM 0: AM 1: PM	R
7	—	The read value is undefined.	R

Note: S-TYPE3, P-TYPE3

RHRCP_n is a read-only register that captures the RHRCNT value when a time capture event is detected.

The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the RHRCP0, RHRCP1, and RHRCP2 registers, respectively.

The PM bit is only enabled when the RCR2.HR24 bit is 0 (in 12-hour mode).

This register is cleared to 0x00 by an RTC software reset. Before reading from this register, you must stop the time capture event detection using the RTCCR_n.TCCT[1:0] bits.

27.2.31 RDAYCP_n : Date Capture Register n (n = 0 to 2) (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x5A + 0x10 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	DATE10[1:0]	DATE1[3:0]				
Value after reset:	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
3:0	DATE1[3:0]	1-Day Capture Capture value for the ones place of days.	R
5:4	DATE10[1:0]	10-Day Capture Capture value for the tens place of days.	R
7:6	—	The read value is undefined.	R

Note: S-TYPE3, P-TYPE3

RDAYCP_n is a read-only register that captures the RDAYCNT value when a time capture event is detected.

The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the RDAYCP0, RDAYCP1, and RDAYCP2 registers, respectively.

This register is cleared to 0x00 by an RTC software reset. Before reading from this register, the time capture event detection should be stopped using the RTCCRn.TCCT[1:0] bits.

27.2.32 RMONCPn : Month Capture Register n (n = 0 to 2) (in Calendar Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x5C + 0x10 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	MON1 0	MON1[3:0]			
Value after reset:	0	0	0	x	x	x	x	x

Bit	Symbol	Function	R/W
3:0	MON1[3:0]	1-Month Capture Capture value for the ones place of months.	R
4	MON10	10-Month Capture Capture value for the tens place of months.	R
7:5	—	These bits are read as 0.	R

Note: S-TYPE3, P-TYPE3

RMONCPn is a read-only register that captures the RMONCNT value when a time capture event is detected.

The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the RMONCP0, RMONCP1, and RMONCP2 registers, respectively.

This register is cleared to 0x00 by an RTC software reset. Before reading from this register, the time capture event detection should be stopped using the RTCCRn.TCCT[1:0] bits.

27.2.33 BCNTnCPm : BCNTn Capture Register m (n= 0 to 3, m = 0 to 2) (in Binary Count Mode)

Base address: RTC = 0x4020_2000
RTC_NS = 0x5020_2000

Offset address: 0x52 + 0x10 × m (BCNT0CPm)
0x54 + 0x10 × m (BCNT1CPm)
0x56 + 0x10 × m (BCNT2CPm)
0x5A + 0x10 × m (BCNT3CPm)

Bit position:	7	6	5	4	3	2	1	0
Bit field:								
Value after reset:	x	x	x	x	x	x	x	x

BCNTnCPm is a read-only register that captures the BCNTn value when a time capture event is detected. BCNT3CPm is assigned to the BCNTCPm[31:24] bits, BCNT2CPm is assigned to the BCNTCPm[23:16] bits, BCNT1CPm is assigned to the BCNTCPm[15:8] bits and BCNT0CPm is assigned to the BCNTCPm[7:0] bits. The event detection times detected by the RTCIC0, RTCIC1, and RTCIC2 pins are stored in the BCNTnCP0, BCNTnCP1, and BCNTnCP2 registers, respectively.

This register is cleared to 0x00 by an RTC software reset. Before reading from this register, you must stop the time capture event detection using the RTCCRn.TCCT[1:0] bits.

27.3 Operation

27.3.1 Outline of Initial Settings of Registers After Power On

After the power is turned on, perform the initial settings for the clock, count mode, time error adjustment, time, alarm, interrupts, and time capture.

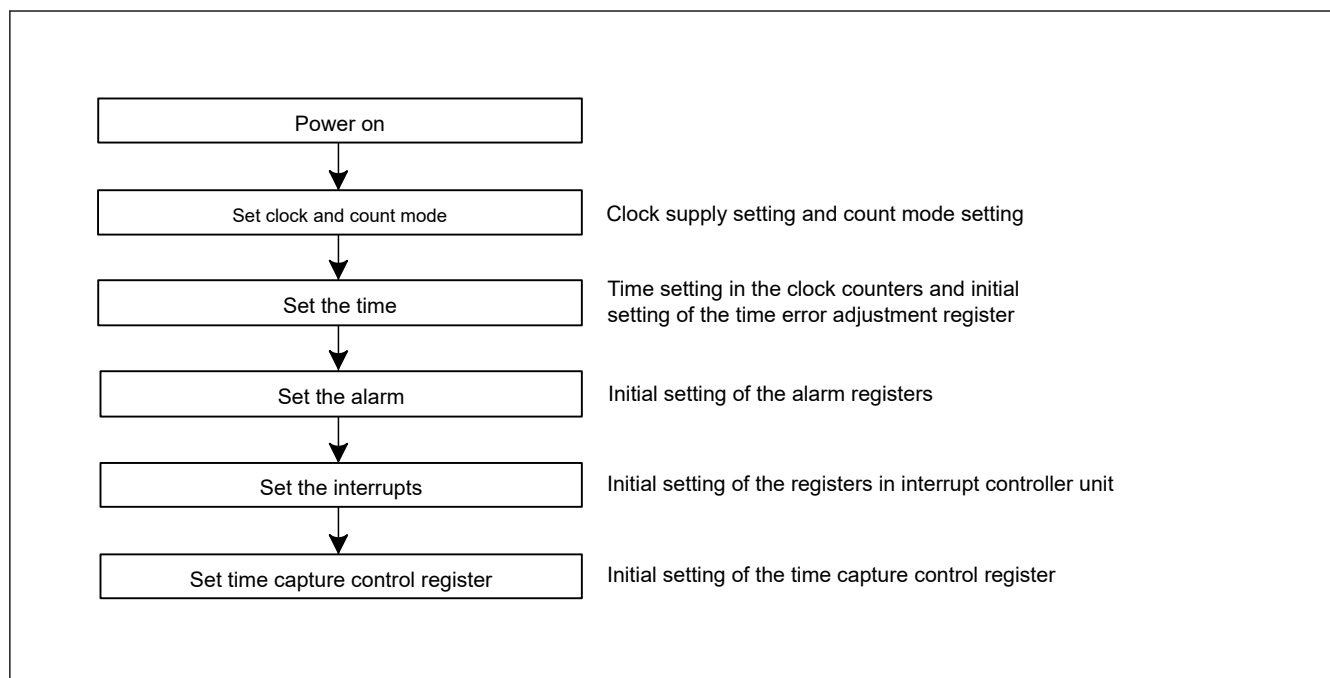


Figure 27.2 Outline of initial settings after a power on

27.3.2 Clock and Count Mode Setting Procedure

[Figure 27.3](#) shows how to set the clock and the count mode.

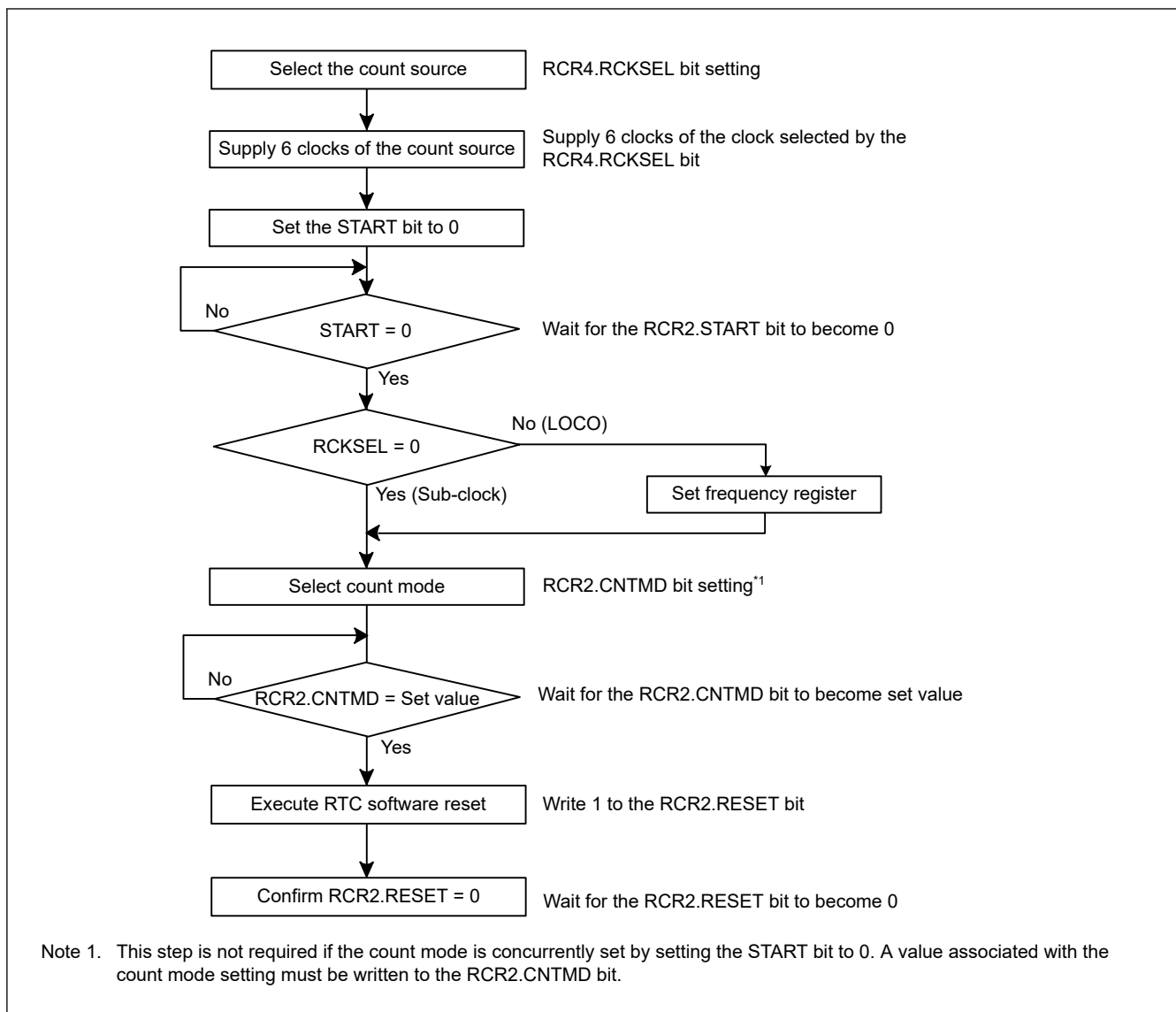


Figure 27.3 Clock and count mode setting procedure

27.3.3 Setting the Time

Figure 27.4 shows how to set the time.

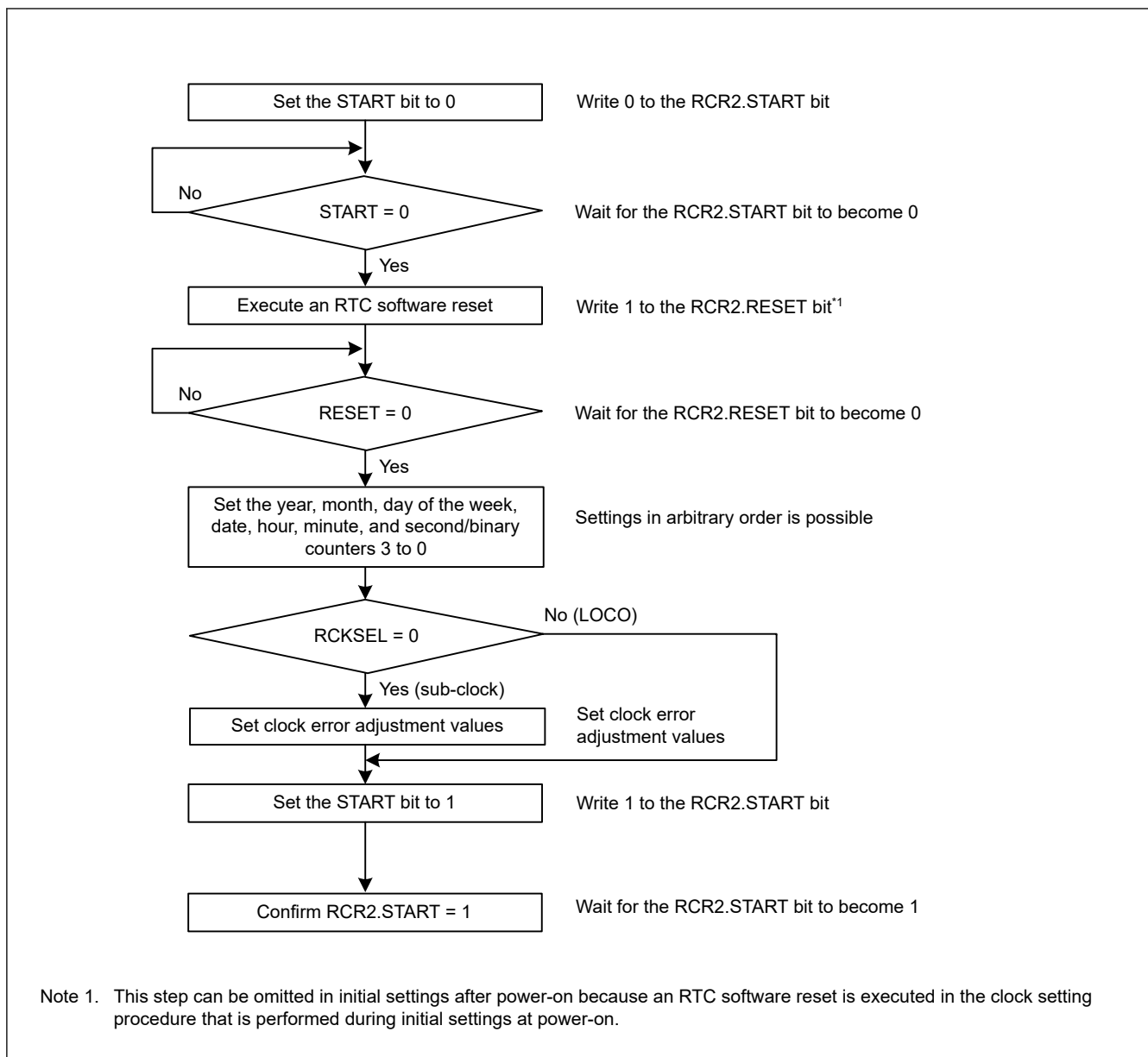


Figure 27.4 Setting the time

27.3.4 30-Second Adjustment

Figure 27.5 shows how to execute a 30-second adjustment.

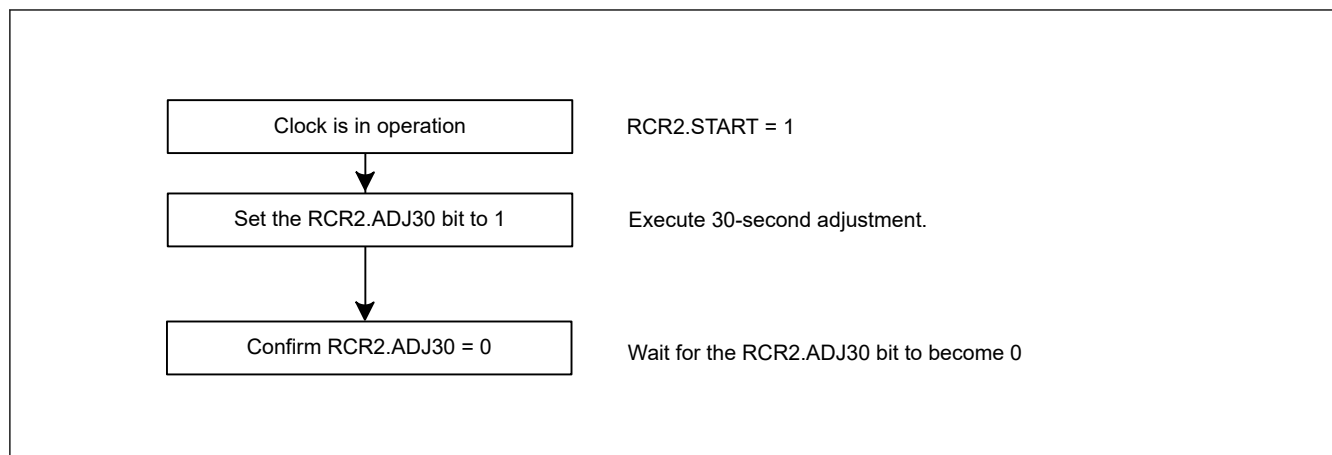
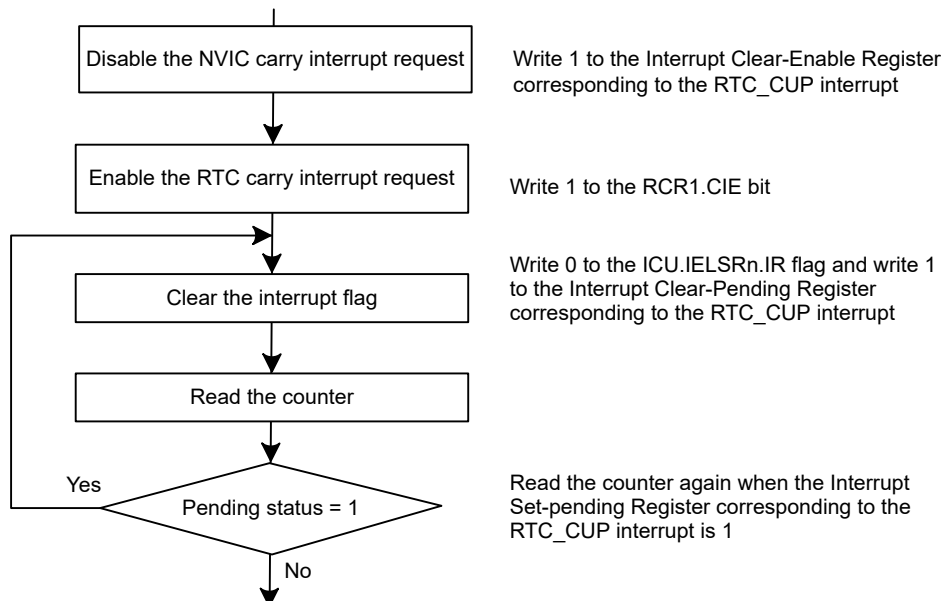


Figure 27.5 30-second adjustment

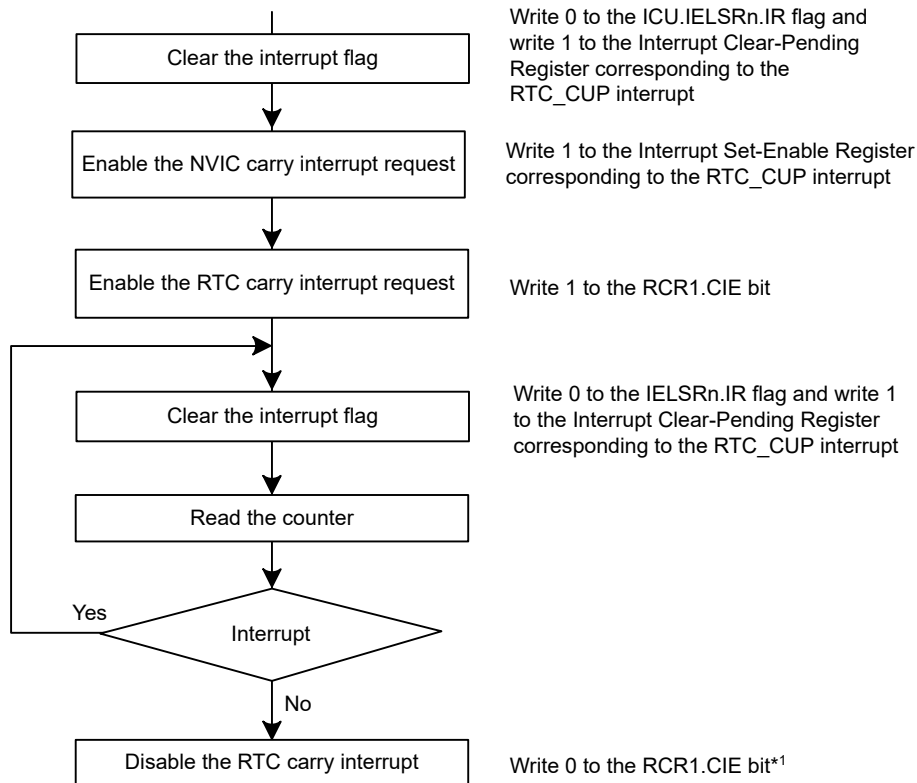
27.3.5 Reading 64-Hz Counter and Time

[Figure 27.6](#) shows how to read a 64-Hz counter and time.

(a) To read the time without using interrupt



(b) To read the time using interrupts



Note 1. Disable interrupts if required.

Figure 27.6 Reading time

If a carry occurs while the 64-Hz counter and time are read, the correct time is not obtained, therefore they must be read again. The procedure for reading the time without using interrupts is shown in (a) in Figure 27.6, and the procedure using carry interrupts is shown in (b). To keep the program simple, method (a) should be used in most cases.

27.3.6 Alarm Function

Figure 27.7 shows how to use the alarm function.

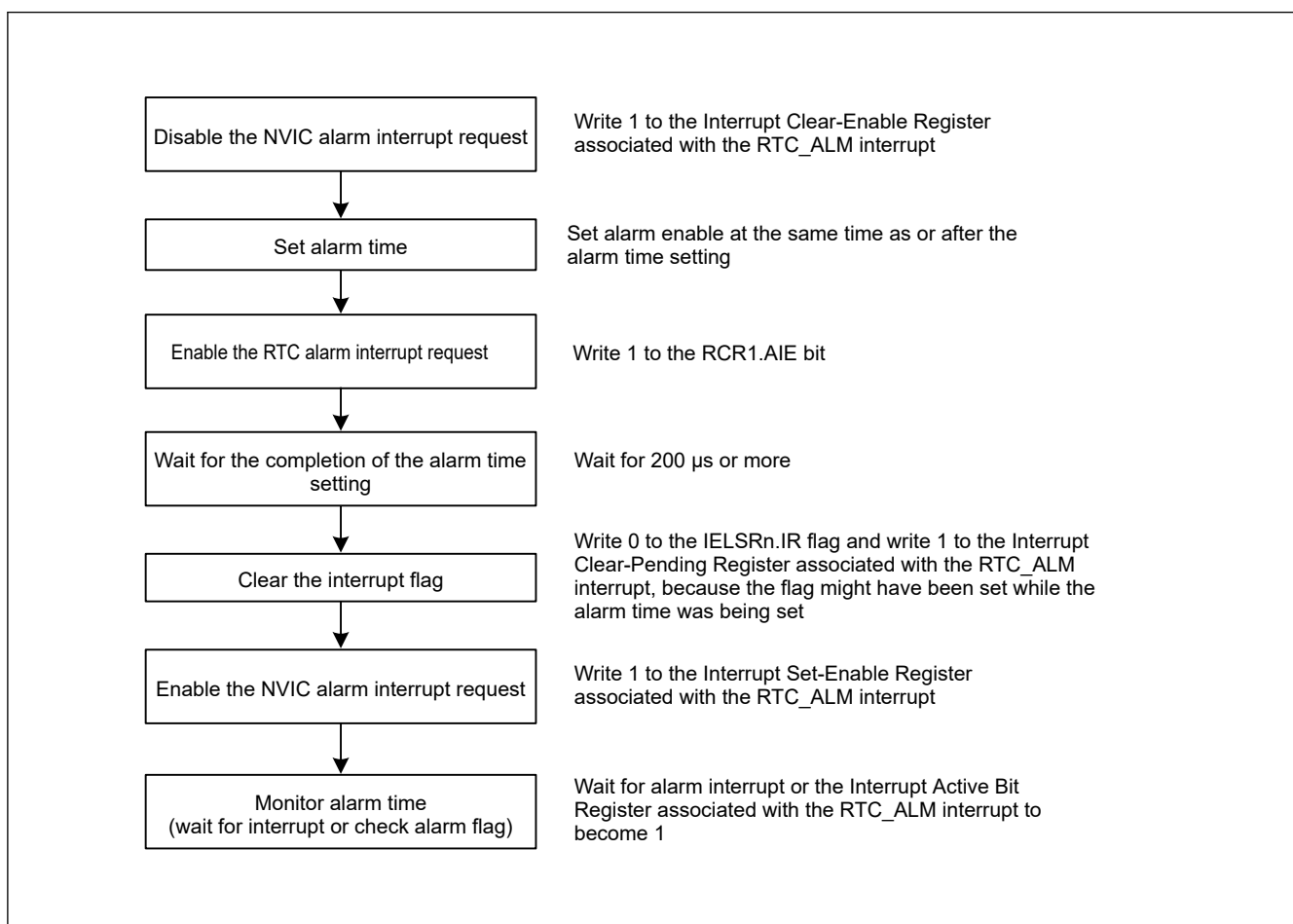


Figure 27.7 Using the alarm function

In calendar count mode, an alarm can be generated by any one of year, month, date, day-of-week, hour, minute or second, or any combination of those. Write 1 to the ENB bit in the alarm registers involved in the alarm setting, and set the alarm time in the lower bits. Write 0 to the ENB bit in registers not involved in the alarm setting.

In binary count mode, an alarm can be generated in any bit combination of 32 bits. Write 1 to the ENB bit of the Alarm Enable register associated with the target bit of the alarm, and set the alarm time in the alarm register. For bits that are not the target of the alarm, write 0 to the ENB bit of the Alarm Enable register.*¹

For any of the ENB[31:0] bits that are set to 1, the bits in the corresponding positions in the binary counter (BCNT[31:0]) are compared with the values of the corresponding bits in the binary alarm registers*¹. When all such bits match, the IR flag associated with the RTC_ALM interrupt is set to 1 and the corresponding bits in the Interrupt Set-Pending/Clear-Pending Registers are set to 1. Alarm detection can be confirmed by reading the Interrupt Set-Pending Register associated with the RTC_ALM interrupt, but an interrupt should be used in most cases. If 1 is set in the Interrupt Set-Enable Register associated with the RTC_ALM interrupt, an alarm interrupt is generated in the event of the alarm, enabling the alarm to be detected.

Writing 0 sets the IELSRn.IR flag associated with the RTC_ALM interrupt to 0. If interrupt is enabled, the Interrupt Set-Pending/Clear-Pending Register associated with the RTC_ALM interrupt is cleared automatically after exiting the interrupt handler. Otherwise, write 1 to the Interrupt Clear-Pending Register associated with the RTC_ALM interrupt to clear it.

When the counter and the alarm time match in a low power state, the MCU returns from the low power state.

Note 1. For any bits in the ENB bits that are set to 1, the values in the corresponding positions in the alarm registers from the following registers are compared with the corresponding bits of the counted values.

Counter registers: RSECCNT, RMINCNT, RHRCNT, RWKCNT, RDAYCNT, RMONCNT, RYRCNT

Alarm registers: RSECAR, RMINAR, RHRAR, RWKAR, RDAYAR, RMONAR, RYRAREN

27.3.7 Procedure for Disabling Alarm Interrupt

Figure 27.8 shows the procedure for disabling the enabled alarm interrupt request.

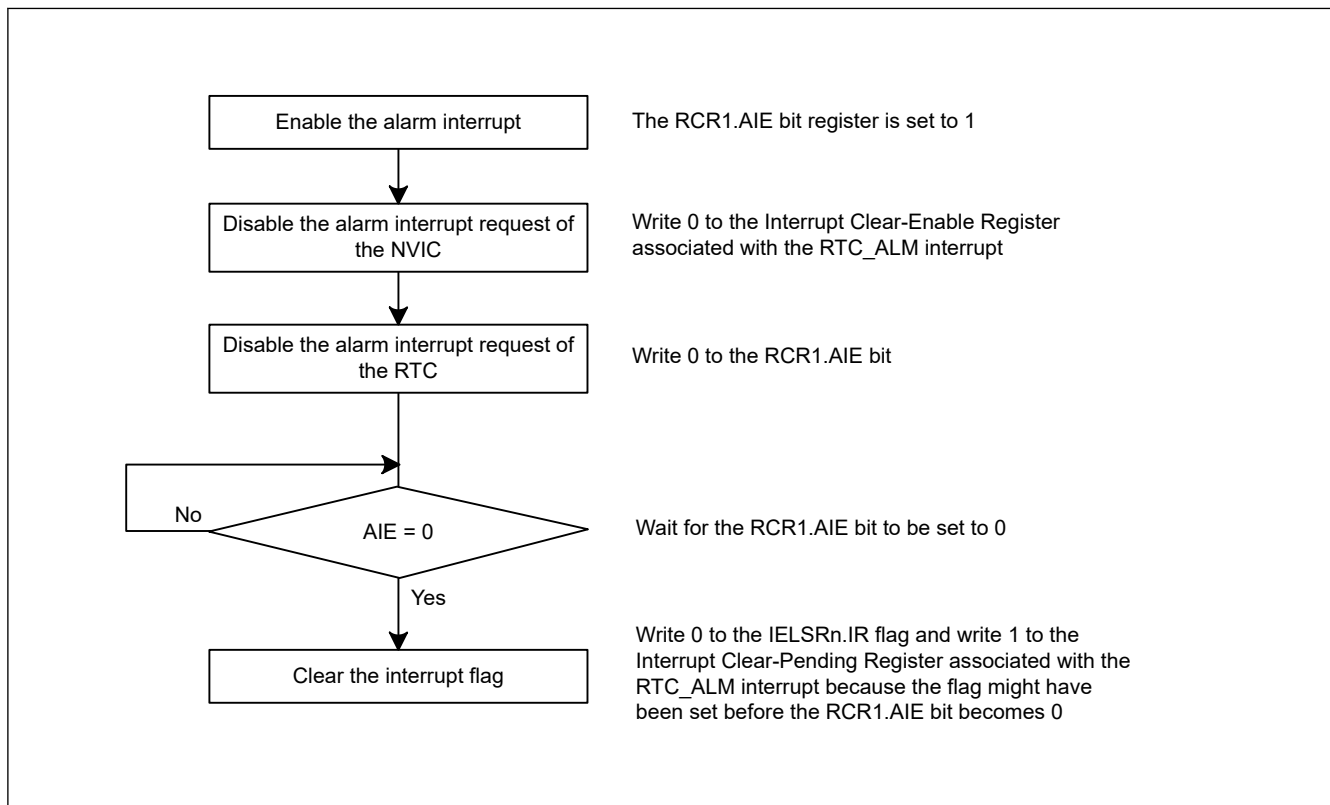


Figure 27.8 Procedure for disabling alarm interrupt request

27.3.8 Time Error Adjustment Function

The time error adjustment function is used to correct errors, running fast or slow, in the time caused by variation in the precision of oscillation by the sub-clock oscillator. Because 32768 cycles of the sub-clock oscillator constitute 1 second of operation when the sub-clock oscillator is selected, the clock runs fast if the sub-clock frequency is high and slow if the sub-clock frequency is low.

The time error adjustment functions include:

- Automatic adjustment
- Adjustment by software

Use the RCR2.AADJE bit to select automatic adjustment or adjustment by software.

27.3.8.1 Automatic Adjustment

Enable automatic adjustment by setting the RCR2.AADJE bit to 1.

Automatic adjustment is the addition or subtraction of the value counted by the prescaler to or from the value in the RADJ register every time the adjustment period selected by the RCR2.AADJP bit elapses.

(1) Example 1: Sub-clock oscillator running at 32.769 kHz**Adjustment procedure**

When the sub-clock oscillator is running at 32.769 kHz, 1 second elapses every 32769 clock cycles. The RTC is meant to run at 32768 clock cycles, so the clock runs fast by 1 clock cycle every second. The time on the clock is fast by 60 clock cycles per minute, so adjustment can take the form of setting the clock back by 60 cycles every minute.

Register settings when RCR2.CNTMD = 0:

- RCR2.AADJP = 0 (adjustment every minute)
- RADJ.PMADJ[1:0] = 10b (adjustment is performed by the subtraction from the prescaler)
- RADJ.ADJ[5:0] = 60 (0x3C)

(2) Example 2: Sub-clock oscillator running at 32.766 kHz**Adjustment procedure**

When the sub-clock oscillator is running at 32.766 kHz, 1 second elapses every 32766 clock cycles. The RTC is meant to run at 32768 clock cycles, so the clock runs slow by 2 clock cycles every second. The time on the clock is slow by 20 clock cycles every 10 seconds, so adjustment can take the form of setting the clock forward by 20 cycles every 10 seconds.

Register settings when RCR2.CNTMD = 0:

- RCR2.AADJP = 1 (adjustment every 10 seconds)
- RADJ.PMADJ[1:0] = 01b (adjustment is performed by the addition to the prescaler)
- RADJ.ADJ[5:0] = 20 (0x14)

(3) Example 3: Sub-clock oscillator running at 32.764 kHz**Adjustment procedure**

When the sub-clock oscillator is running at 32.764 kHz, 1 second elapses on 32764 clock cycles. Because the RTC operates for 32768 clock cycles as 1 second, the clock is delayed for 4 clock cycles per second. In 8 seconds, the delay is 32 clock cycles, therefore correction can be made by advancing the clock 32 clock cycles every 8 seconds.

Register settings when RCR2.CNTMD = 1:

- RCR2.AADJP = 1 (adjustment every 8 seconds)
- RADJ.PMADJ[1:0] = 01b (adjustment is performed by the addition to the prescaler)
- RADJ.ADJ[5:0] = 32 (0x20)

27.3.8.2 Adjustment by Software

Enable adjustment by software by setting the RCR2.AADJE bit to 0. Adjustment by software is the addition or subtraction of the value counted by the prescaler to or from the value in the RADJ register on execution of a write instruction to the RADJ register.

(1) Example 1: Sub-clock oscillator running at 32.769 kHz**Adjustment procedure**

When the sub-clock oscillator is running at 32.769 kHz, 1 second elapses every 32769 clock cycles. The RTC is meant to run at 32768 clock cycles, so the clock runs fast by 1 clock cycle every second. The time on the clock is fast by 1 clock cycle per second, so adjustment can take the form of setting the clock back by 1 cycle every second.

Register settings

- RADJ.PMADJ[1:0] = 10b (adjustment is performed by the subtraction from the prescaler)
- RADJ.ADJ[5:0] = 1 (0x01)

This is written to the RADJ register once per 1-second interrupt.

27.3.8.3 Procedure to Change the Mode of Adjustment

When changing the mode of adjustment, change the value of the AADJE bit in RCR2 after setting the RADJ.PMADJ[1:0] bits to 00b (adjustment is not performed).

To change adjustment by software to automatic adjustment:

1. Set the RADJ.PMADJ[1:0] bits to 00b (adjustment is not performed).
2. Set the RCR2.AADJE bit to 1 (automatic adjustment is enabled).
3. Use the RCR2.AADJP bit to select the period of adjustment.
4. In RADJ, set the PMADJ[1:0] bits for addition or subtraction and the ADJ[5:0] bits to the value for use in time error adjustment.

To change automatic adjustment to adjustment by software:

1. Set the RADJ.PMADJ[1:0] bits to 00b (adjustment is not performed).
2. Set the RCR2.AADJE bit to 0 (adjustment by software is enabled).
3. Proceed with the adjustment by setting the RADJ.PMADJ[1:0] bits for addition or subtraction and the RADJ.ADJ[5:0] bits to the value for use in time error adjustment at the wanted time. After that, the time is adjusted every time a value is written to the RADJ register.

27.3.8.4 Procedure to Stop Adjustment

Stop the adjustment by setting the RADJ.PMADJ[1:0] bits to 00b (adjustment is not performed).

27.3.9 Capturing the Time

The RTC is capable of storing the month, date, hour, minute and second/binary counters 3 to 0 by detecting an edge of a signal on a time capture event input pin in calendar count mode or binary count mode.

A noise filter can also be used on a time capture event input pin. If the noise filter is enabled, the RTCCRn.TCST bit is set to 1 when the input level on the pin matches three times.

The noise filter can be switched on or off for each of the time capture event input pins. Set VBTICTLR.VCHnIEN (n = 0 to 2) to 1 to enable the RTCICn input. Operation when the noise filter is off is shown in [Figure 27.9](#) and operation when the noise filter is on is shown in [Figure 27.10](#).

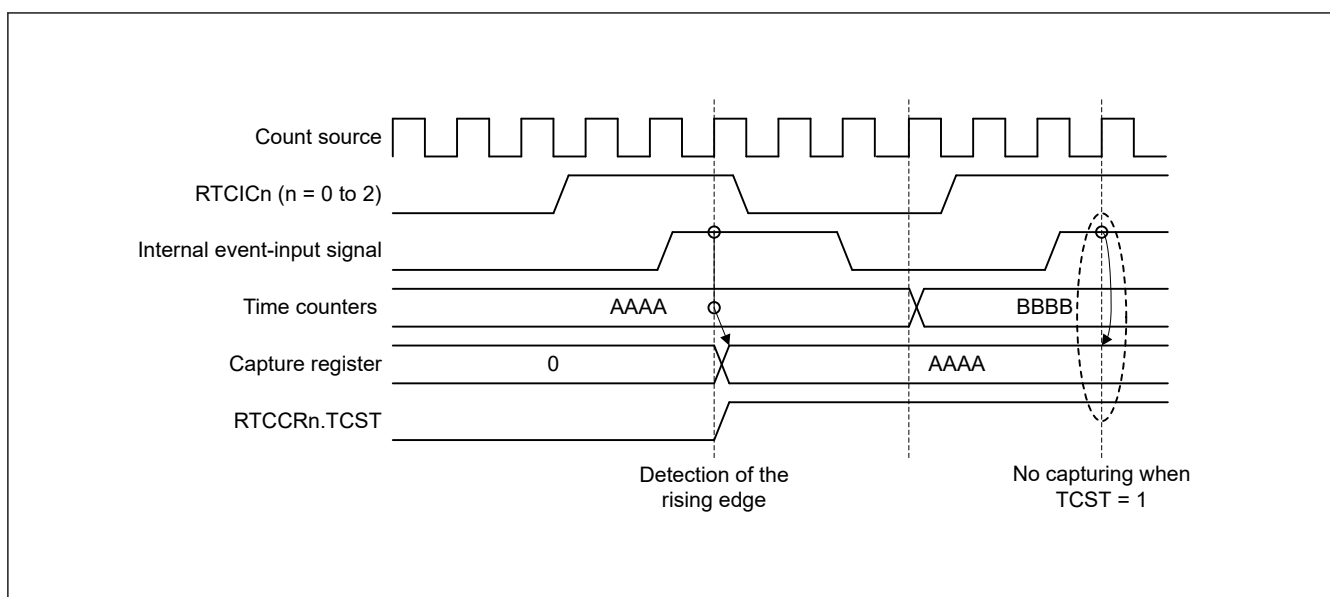


Figure 27.9 Timing of a time capture operation with the noise filter off

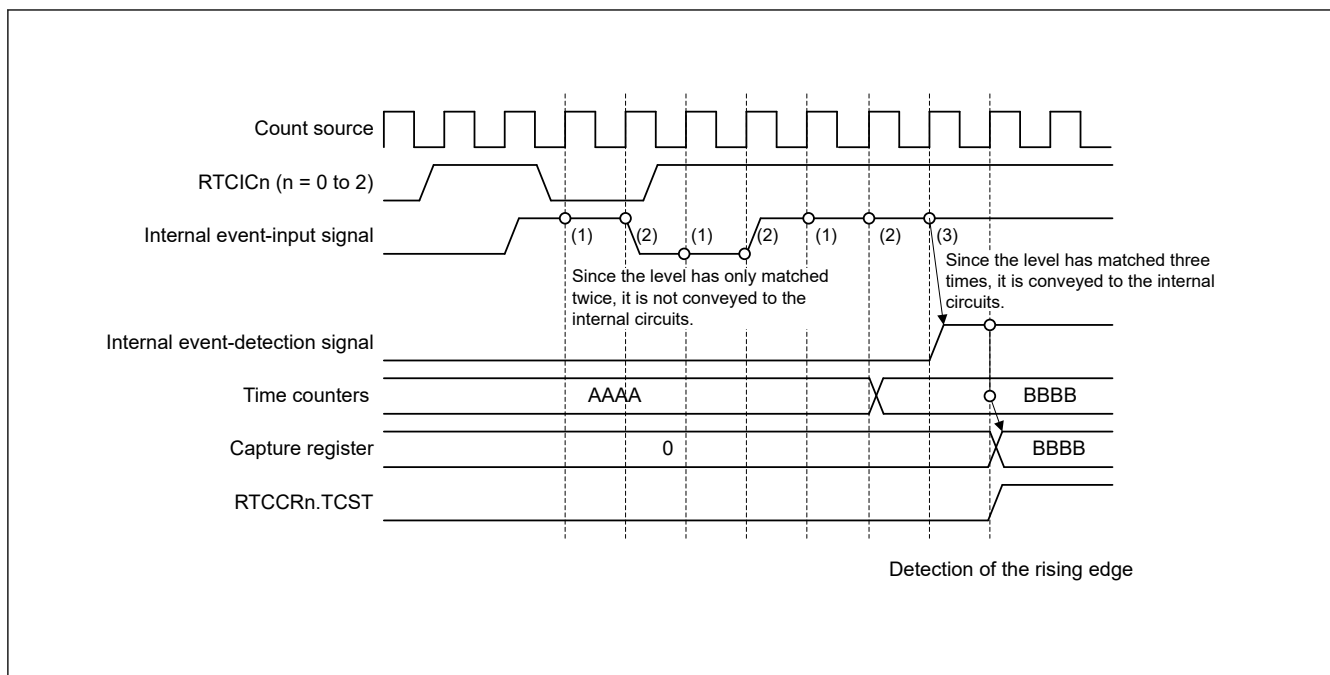


Figure 27.10 Timing of a time capture operation with the noise filter on

27.4 Interrupt Sources

The RTC has three interrupt sources, as listed in [Table 27.3](#).

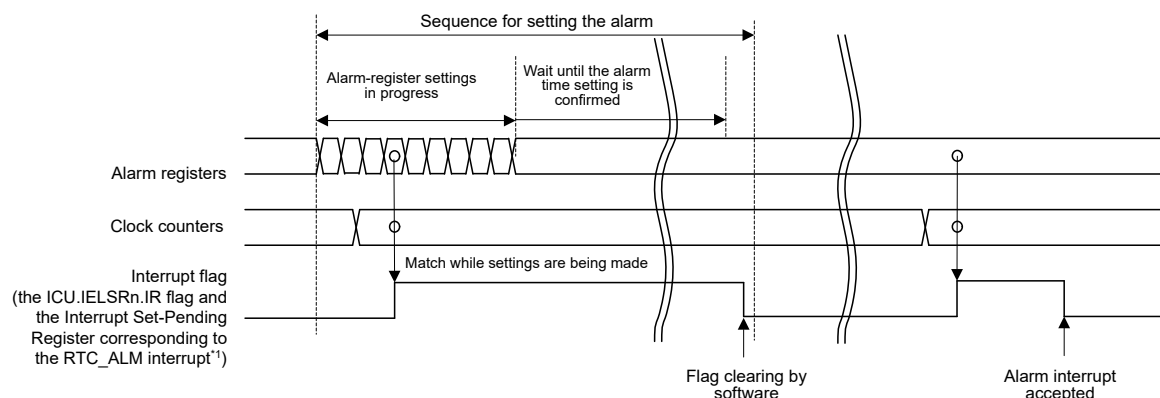
Table 27.3 RTC interrupt sources

Name	Interrupt source
RTC_ALM	Alarm interrupt
RTC_PRD	Periodic interrupt
RTC_CUP	Carry interrupt

(1) Alarm interrupt (RTC_ALM)

This interrupt is generated based on the comparison result between the alarm registers and RTC counters. For details, see [section 27.3.6. Alarm Function](#).

Because there is a possibility that the interrupt flag might be set to 1 when the settings of the alarm registers match the clock counters, wait for the alarm time settings to be confirmed and clear the IELSRn.IR flag and the interrupt Set-Pending Register associated with the RTC_ALM interrupt to 0 again after modifying values of the alarm registers. After the interrupt flag for the alarm interrupt is set to 1 and the state is returned to mismatching of the alarm registers and clock counters, the flag is not 1 again until there is another match or the values of the alarm registers are modified again.



Note 1. See [section 14, Interrupt Controller Unit \(ICU\)](#) for details on the associated interrupt vector number.

Figure 27.11 Timing for the alarm interrupt (RTC_ALM)

(2) Periodic interrupt (RTC_PRD)

This interrupt is generated at intervals of 2 seconds, 1 second, 1/2 second, 1/4 second, 1/8 second, 1/16 second, 1/32 second, 1/64 second, 1/128 second, or 1/256 second. The interrupt interval can be selected in the RCR1.PES[3:0] bits.

(3) Carry interrupt (RTC_CUP)

This interrupt is generated when a carry to the second counter/binary counter 0 occurred or a carry to the R64CNT counter occurred during read access to the 64-Hz counter.

[Figure 27.12](#) shows the timing of the carry interrupt (RTC_CUP).

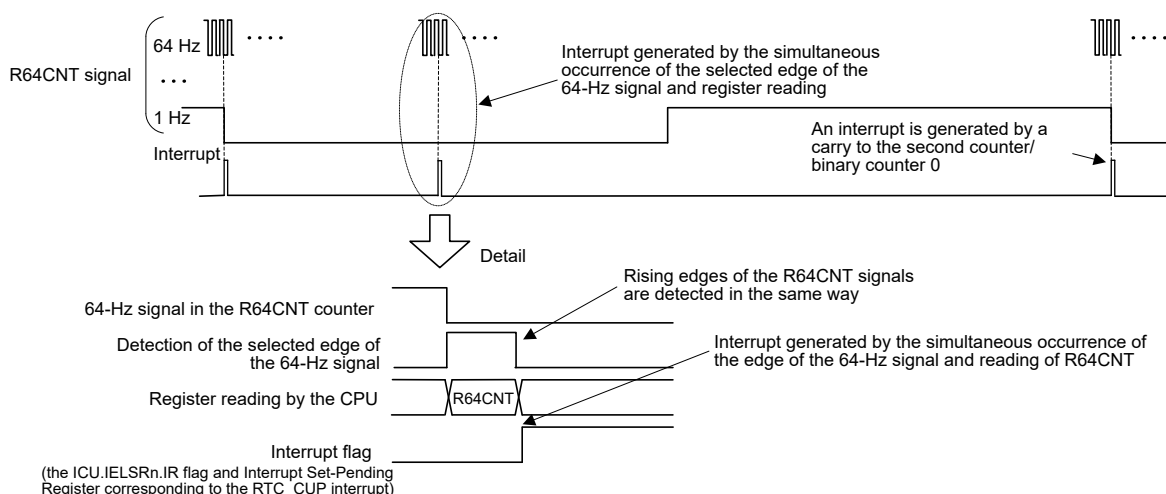


Figure 27.12 Timing for the carry interrupt (RTC_CUP)

27.5 Event Link Output

The RTC generates periodic event output (RTC_PRD) event signal for the ELC that can be used to initiate operations by other modules selected in advance.

The periodic event signal is output at the interval selected from 1/256, 1/128, 1/64, 1/32, 1/16, 1/8, 1/4, 1/2, 1, and 2 seconds by setting the RCR1.PES[3:0] bits.

The event generation period immediately after the event generation is selected is not guaranteed.

Note: If event linking from the RTC is used, only set the ELC after setting the RTC, for example initialization and time settings. Setting the RTC after the ELC can lead to output of unexpected event signals.

27.5.1 Interrupt Handling and Event Linking

The RTC has a bit to enable or disable periodic interrupts. An interrupt request signal is output to the CPU when an interrupt source is generated while the associated enable bit is enabled.

In contrast, an event link output signal is sent to other modules as an event signal through the ELC when an interrupt source is generated, regardless of the setting of the associated interrupt enable bit.

Note: Although alarm and periodic interrupts can still be output during Software Standby or Deep Software Standby mode, the periodic event signals for the ELC are not output.

27.6 Usage Notes

27.6.1 Register Writing During Counting

The following registers should not be written to during counting, that is, while the RCR2.START bit is 1:

- RSECCNT/BCNT0
- RMINCNT/BCNT1
- RHRCNT/BCNT2
- RDAYCNT
- RWKCNT/BCNT3
- RMONCNT
- RYRCNT
- RCR1.RTCOS
- RCR2.RTCOE
- RCR2.HR24
- RFRL

The counter should be stopped before writing to any of these registers.

27.6.2 Use of Periodic Interrupts

Figure 27.13 shows the procedure for using periodic interrupts.

The generation and period of the periodic interrupt can be changed by setting the RCR1.PES[3:0] bits. However, because the prescaler R64CNT and RSECCNT/BCNT0 are used to generate interrupts, the interrupt period is not guaranteed immediately after setting the RCR1.PES[3:0] bits.

In addition, any of the following operation can affect the interrupt period:

- Stopping/restarting or resetting counter operation
- Reset by RTC software
- 30-second adjustment by changing the RCR2 value

When the time error adjustment function is used, the interrupt generation period after adjustment is added or subtracted based on the adjustment value.

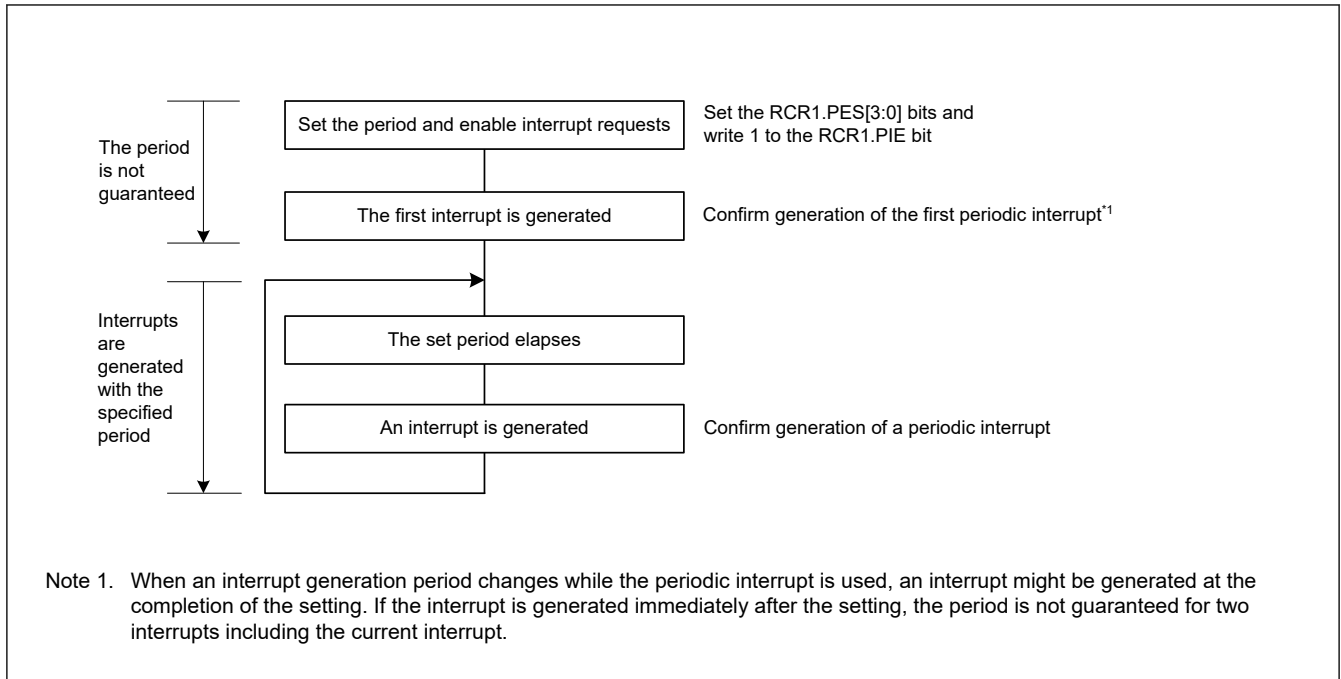


Figure 27.13 Using the periodic interrupt function

27.6.3 RTCOUT (1-Hz/64-Hz) Clock Output

Stopping/restarting or resetting counter operation, reset by RTC software, and the 30-second adjustment by changing the RCR2 value affects the period of RTCOUT (1-Hz/64-Hz) output. When the time error adjustment function is used, the period of RTCOUT (1-Hz/64-Hz) output after adjustment is added or subtracted based on the adjustment value.

27.6.4 Transitions to Low Power Modes After Setting Registers

A transition to a low power state (Software Standby mode, Deep Software Standby mode, or battery backup state) during a write to an RTC register might corrupt the value of the register. After setting the register, confirm that the setting is in place before initiating a transition to a low power state.

27.6.5 Notes on Writing to and Reading from Registers

- When reading a counter register such as the second counter after writing to the counter register, follow the procedure in [section 27.3.5. Reading 64-Hz Counter and Time](#).
- The value written to the count registers, alarm registers, year alarm enable register, bits RCR2.AADJE, AADJP, and HR24, RCR4 register, or frequency register is reflected when fourth read operations are performed after writing.
- The values written to the RCR1.CIE, RCR1.RTCOS, and RCR2.RTCOE bits can be read immediately after writing.
- To read the value from the timer counter after returning from a reset or a period in Software Standby mode, Deep Software Standby mode, or battery backup state, wait for 1/128 second while the clock is operating (RCR2.START bit = 1).
- After a reset is generated, write to the RTC register after 6 cycles of the count source clock have elapsed.

27.6.6 Changing the Count Mode

When changing the count mode (calendar count mode/binary count mode), set the RCR2.START bit to 0, stop the counting operation, then start it again from the initial setting. For details on the initial setting, see [section 27.3.1. Outline of Initial Settings of Registers After Power On](#).

27.6.7 Initialization Procedure When the RTC Is Not to Be Used

Registers in the RTC are not initialized by a reset. Depending on the initial state, the generation of an unintentional interrupt request or operation of the counter might lead to increased power consumption.

For applications that do not require a realtime clock, initialize the registers by following the initialization procedure shown in [Figure 27.14](#).

Alternatively, when the sub-clock oscillator is not used as the system clock source, realtime clock, operating clock of each peripheral modules or FLL function of HOCO, the counter can be stopped by writing 0 (sub-clock oscillator is selected) to the RCR4.RCKSEL bit and stopping the sub-clock oscillator. To stop the sub-clock oscillator, write 1 to the SOSCCR.SOSTP bit.

For details on the setting of the SOSCCR.SOSTP bit, see [section 9, Clock Generation Circuit](#).

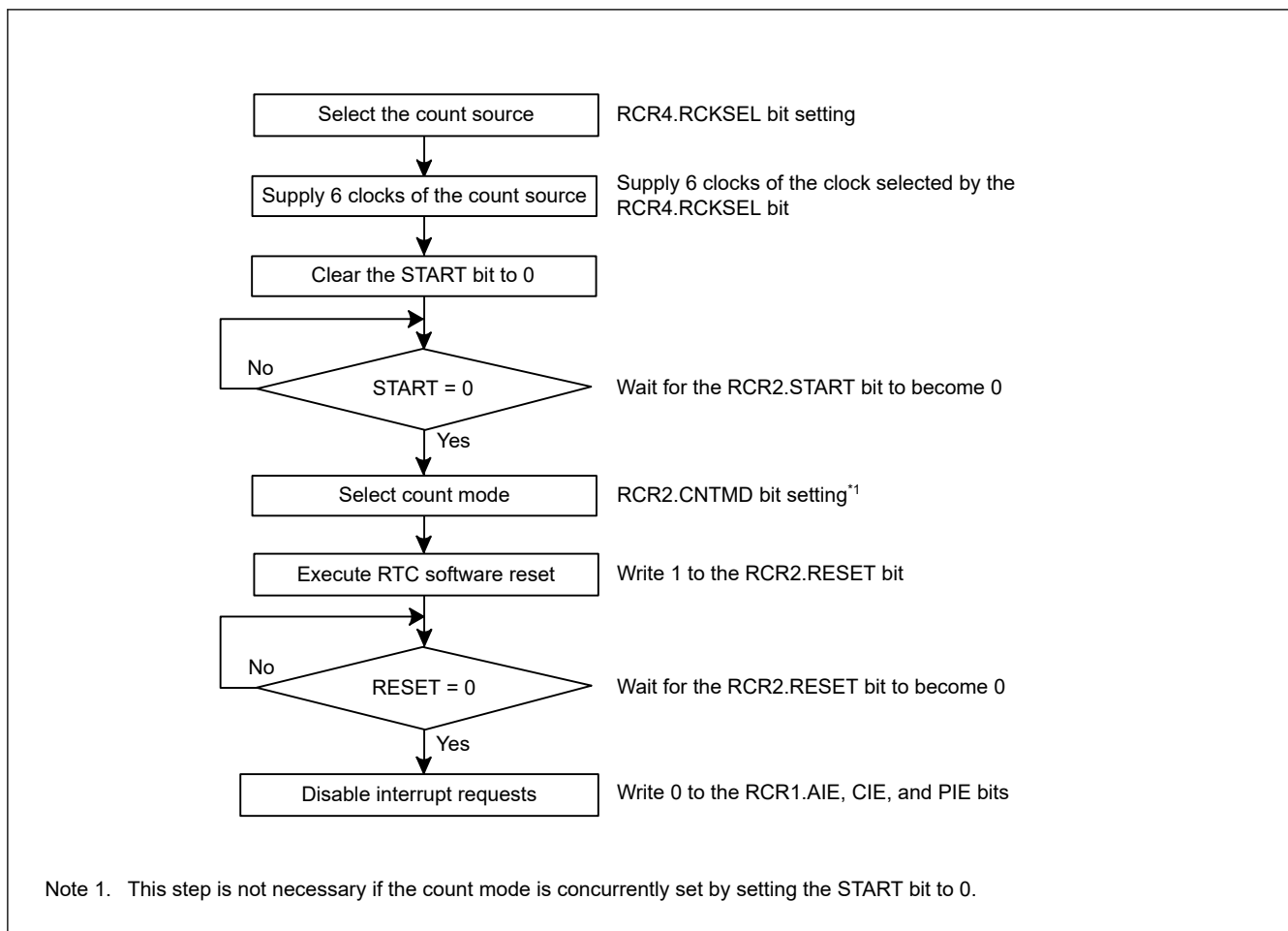


Figure 27.14 Initialization procedure

27.6.8 When Switching Source Clock

When switching a clock source by changing SCKSCR.CKSEL[2:0], the clock output from the selector stops for 4 cycles of the switched clock. If the RTC periodical interrupt or RTC periodical event output was generated at this time, the interrupt or event is invalid.

27.6.9 When Writing to the RTC Registers

If VCC voltage of the MCU is lower than 1.8V and write access to the RTC registers is performed, read access should be performed at least once after one write access is performed, or for the consecutive write access, leave an interval of 167ns for each access.